Experimental Evidence on the Complementarity of Union Membership

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June 2026

Abstract

This paper examines whether an individual's union membership status is affected by the membership rate in their workplace. In an information provision experiment, I randomly provided teachers with an estimate of their school district's unionization rate. Treated teachers updated their beliefs toward the provided estimate and adjusted their planned union participation accordingly: a 10 percentage point change in beliefs led to a 3.6 percentage point shift in planned participation. Effects on actual membership status one year later are muted overall, but pronounced in highly unionized districts. This implies that union membership exhibits complementarity, especially in workplaces with strong union support.

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I. Introduction

Collective action issues arise when a group or organization provides a public good but individuals bear the cost of participation. Without compulsory mechanisms in place, individuals have an incentive to free ride even if they support and benefit from the collective good (Olson 1965). As a result, the provision of the public good may unravel if everyone acts in their self-interest, with important implications in contexts ranging from political organizing to labor relations.

This is especially relevant in the case of union membership given that many unionized workplaces cannot mandate that all workers pay dues. In the US, about half the states have right-to-work laws prohibiting compulsory union dues and the Supreme Court has ruled that mandatory dues in the public sector violate the First Amendment (Janus v. AFSCME 2018). Yet despite the recent Supreme Court case, about two-thirds of teachers say that they are union members (National Center for Education Statistics 2021).

This raises the question of how unions sustain membership rates in the face of the free-rider problem. One explanation is the role of social customs: workers who join the union may gain a reputational benefit from their colleagues (Booth 1985, Naylor and Cripps 1993). Despite its importance, there is little empirical work that studies whether social mechanisms — in particular the participation of others — affect individual membership decisions.

There are two key challenges in estimating whether group membership rates, such as within a firm or school district, affect individual membership decisions. First, it requires individual-level data on union membership within specific organizations. Second, even with detailed membership data, it is difficult to isolate exogenous group shocks that do not also affect an individual by definition. For instance, quasi-experimental variation in co-workers' membership status, such as from a policy change, is also likely to affect an individual's membership choice, an example of a reflection problem (Manski 1993).

To overcome these empirical issues, I conduct an information provision experiment with a sample of teachers in Wisconsin to understand whether perceptions of others' union participation affect one's own participation. My study combines administrative data on individual union membership with an online survey that elicits prior and posterior beliefs. I measure individual union membership status by linking public-school personnel data with campaign finance reports that reveal which teachers pay union dues. In the survey, I first elicit prior beliefs by asking teachers to predict the membership rate in their school district. Measuring prior beliefs is crucial because the information treatment could have offsetting effects if some teachers update positively and others update negatively. Half of the sample is then randomly assigned to receiving a truthful estimate of their school district's union membership rate. After the information treatment, I elicit posterior beliefs by asking all participants to predict next year's membership rate. This allows me to see if teachers have misperceptions about the average membership rate in their district, whether they correct those misperceptions in response to the treatment, and whether a change in beliefs ultimately affects the likelihood of being a member themselves.

Theoretically, a change in one's belief about the group membership rate could affect one's own membership decision in either direction. On the one hand, a teacher who learns that district membership is higher than they thought may become more likely to join the union. This aligns with the social customs model, which predicts that individual and group participation decisions are complementary. Likewise, a teacher may update on the quality of the union or the services that the union provides. Conversely, recent experimental work on protest participation and election campaigning finds evidence of strategic substitutes (Cantoni et al. 2019; Jarke-Neuert, Perino, and Schwickert 2023; Hager et al. 2023). In these settings, people were *less likely* to participate when more members of a group participated, consistent with the logic of free-riding.

Almost 2,500 teachers from 290 school districts participated in the experimental survey during the spring of the 2023–24 school year. Importantly, there is considerable variation in prior beliefs both below and above the estimated district membership rates. Therefore, following Cantoni et al. (2019), I estimate a pooled regression, in which treated respondents who underestimated their district rate are coded as 1 and treated respondents who overestimated their district rate are coded as -1. This specification tests whether treated respondents adjusted their beliefs towards the provided district membership estimate (positive coefficients) or away from it (negative coefficients).

The study has five main findings. First, while many teachers accurately predict the membership rate in their district, the mean absolute gap between prior beliefs and the truthful estimate is 17 percentage points. This implies that there is considerable variation in prior beliefs and room for correcting misperceptions. Second, the information provision induced a strong first stage — teachers in the treatment group are significantly more likely to update their posterior beliefs in the direction of the provided estimate. On average, treated teachers reduce the gap between their prior beliefs and the truthful estimate by about 9 percentage points. Third, I measure respondents' "planned participation" by asking them what the likelihood is that they will be a union member next year. The treatment caused respondents to change their planned participation in the same direction as their revised beliefs. In other words, respondents who overestimated the district rate were less likely to say that they would be a member next year, while respondents who underestimated the district rate were more likely to say they would be a member next year. This implies that planned union participation is complementary to beliefs about the group participation rate. Fourth, I then measure respondents' actual membership status one year later by again matching the survey participants to the campaign finance reports. This allows me to see whether the treatment affected participants' actual membership over a longer-run time frame using administrative data. As with planned participation, I find that the point estimates for membership status in the following year are positive, consistent with complementarity, but I cannot reject that the effect was zero. Fifth, this null effect on actual participation masks important heterogeneity: I find suggestive evidence of significant effects on actual membership in districts where a majority of teachers were members at baseline and null effects in districts where a majority of teachers were not members at baseline.

In terms of mechanisms, one possibility is that the complementary effects are driven by social factors. For example, teachers who underestimated the district rate may want to join in the future because they do not want to be seen as a free-rider. Likewise, teachers who overestimated the group rate may wish to leave the union because other people are not paying their fair share. If this were the case, effects would likely be more pronounced in districts with more union support at baseline, given that these districts likely have stronger social mechanisms in place. To test this, I take advantage of the considerable variation in the share of members at the district level. For instance, some districts in my sample have greater than 50 percent membership, making social factors like peer pressure likely more salient. At the same time, many districts have few teachers in the union, which likely makes the information provision less relevant. I therefore split the sample by whether the average district membership rate was below or above 50 percent at baseline.[1] For districts where less than 50 percent of teachers are union members, the point estimate on actual participation a year later is slightly negative and insignificant. However, in districts where a majority are members, my estimated effects are similar in magnitude and significance for both stated intentions and actual participation. While baseline membership rates are inherently not random, this suggests that effects were particularly pronounced in districts that still maintained high levels of union presence, consistent with social customs playing an important role.

In addition to social factors, the complementarity could be driven by respondents updating on the effectiveness of their local union. To disentangle these two mechanisms, I take advantage of information about the survey subjects at baseline. In particular, I have rich information on each individual from the administrative data sources, including whether they were already union members at the time of the experiment and their years of experience in the school district. If respondents update on the quality of the union, then effects would likely be more pronounced for non-members and less experienced teachers, given that members and experienced teachers are likely more informed about the union before the experiment. In support of this idea, I find that members and experienced workers have more accurate priors about the district membership rate and are more likely to correctly identify the union's collective bargaining status. For planned participation, I find similar effects for members and non-members, while for actual membership status, I find suggestive evidence that members responded to the treatment more than non-members. Likewise, I find more pronounced effects on both outcomes for experienced teachers relative to less experienced

[1] This threshold is not merely symbolic as Wisconsin teachers' unions are required to hold annual elections in order to keep their legal standing in the state (Foy 2025).

teachers. While suggestive, these patterns are not consistent with the mechanism of participants updating on union quality.

Lastly, I estimate the results in a two-stage least squares (2SLS) specification, using treatment status as an instrument for the change in posterior beliefs. The 2SLS model is informative because it can be converted to a behavioral elasticity, capturing how an individual's membership status responds to changes in beliefs about others' participation. At one extreme, an elasticity of 1 implies that individual membership rates change one-to-one with the group membership rate. At the other extreme, an elasticity of 0 indicates that an individual's decision is unaffected by changes in the group membership rate. In my 2SLS design, I estimate a coefficient of 0.36 on stated intentions, implying that a 10 percentage point change in beliefs leads to a corresponding 3.6 percentage point change in planned union participation. Given that baseline membership rates and beliefs about others' membership are approximately equal, this coefficient can be roughly interpreted as an elasticity. This indicates that a change in others' union participation (or the perception thereof) has a substantial effect on an individual's intention to participate in the union. In terms of actual participation, I find a positive, insignificant 2SLS coefficient of 0.11. Again, this masks important heterogeneity: in highly unionized districts at baseline, I estimate an elasticity of about 0.50 for both planned and actual participation.

Related literature: This paper contributes to work relating to the determinants of union membership and collective action problems more broadly. In terms of union membership, there is a long literature discussing the free-rider problem in settings where organizations cannot mandate union dues (Olson 1965, Lumsden and Petersen 1975, Ellwood and Fine 1987). Building on Akerlof (1980), Booth (1985) and Naylor and Cripps (1993) theorize that a "social customs" model of unionization can explain why membership rates are positive in open-shop settings. However, to my knowledge, there are no empirical papers that test whether social customs affect individual union membership. I address this gap by studying one specific social mechanism: the relationship between individual membership decisions and perceptions of the group membership rate.

I also build directly on recent experimental work exploring the relationship between individual and group participation in collective action problems. Prior studies have examined this in the context of protest participation (Cantoni et al. 2019; Hager et al. 2022; Jarke-Neuert, Perino, and Schwickert 2023) and election canvassing (Hager et al. 2023).[2] These studies generally find evidence of strategic substitutes in which individuals are less likely to turn out when they believe there is an increase in others' participation. Notably, in contrast to these prior studies, I find evidence of complementarity in union membership.[3] One possible explanation for this difference

[2]In a non-experimental design, González (2020) examines how networks shape the decision to protest in Chile.

[3]Similarly, Naidu (2022), using survey data on essential workers during COVID from Hertel-Fernandez et al.

is that social mechanisms play a greater role in union membership where workers are all part of the same organization. In contrast, free-riding may be easier when it comes to protesting or election participation in large, anonymous groups.

II. Context and Administrative Data

A. Context

The question of how unions overcome free-riding is relevant across many settings because of "right-to-work" laws barring unions from collecting dues from non-members. In the public sector, recent legislation at the state and federal levels has likewise prohibited public-sector labor unions from collecting "fair-share fees" from non-members. For instance, in 2018, the US Supreme Court ruled that no public-sector unions could automatically mandate membership because it violated the First Amendment (Janus v. AFSCME 2018). In my setting, Wisconsin passed similar legislation in 2011 called Act 10 that effectively made the state "right-to-work" for most public-sector workers.

B. Administrative Data

I measure individual union membership by combining personnel data on all Wisconsin public-school teachers with campaign finance data that lists which teachers pay union dues. The personnel data is made publicly available by the Wisconsin Department of Public Instruction and includes a teacher's name, job assignment, pay, and basic demographic characteristics. I link this to publicly available campaign finance reports from the state campaign finance system. In Wisconsin, political action committees (PACs) are required to list all contributors, no matter how small the amount. This is relevant because the state chapter of the National Education Association (NEA) routes a small amount of each member's dues to its PAC.[4] As a result, the teachers' union PAC is required to list each teacher who indirectly contributes to the PAC as a result of paying union dues. I therefore estimate an individual's union membership status by linking the two datasets together by a person's full name and region (see Appendix Section B for more details).

This data linkage serves two purposes in the experimental design. First, I use data from the 2022–23 school year to construct an estimate of each school district's union membership rate. This is the primary information that is provided to the treatment group as described more in the next section. Second, I use the individual membership data both as a pre-experimental baseline

(2020), finds that workers are more likely to state that they would take part in workplace collective action given a higher hypothetical share of coworkers taking that action. The study also has similarities with the literature on conditional cooperation (e.g., see Fischbacher, Gächter, and Fehr (2001)).

[4] About 97 percent of districts in the state are affiliated with the state chapter of the NEA. While other major unions also have PACs, such as the American Federation of Teachers, they do not automatically deduct a small share of membership dues for campaign finance purposes.

control and as a post-experimental outcome. In other words, I examine how the treatment affected future membership decisions, while controlling for an individual's membership status at the time of the experiment.

III. Hypotheses Tested and Experimental Design

A. Descriptive Analysis

The study examines whether individual union participation is influenced by the share of one's colleagues who are members. As a proof of concept using the administrative data, I analyze whether teachers who switch districts are more or less likely to be members based on membership rates in their former and new workplaces. If union participation is solely an individual decision, then moving to a more unionized district should have no effect on their membership status. However, if union membership exhibits complementarity, we would expect someone to be more likely to be a member after moving to a more unionized district.

To test this, I examine simple event study figures of average membership rates around the time that an individual changes districts. I first group districts into quartiles of membership rates in 2017. In Figure 1, I then plot membership rates in event time for teachers who moved to a new district between 2018–20. For simplicity, I plot membership rates for those who moved from the highest and lowest membership districts into any of the four quartiles.[5]

The figures show that teachers who move to less unionized districts are much less likely to be members after moving. For example, the green line in the left panel shows that workers who move from the highest quartile districts to the lowest quartile districts are substantially less likely to be union members by upwards of 30 percentage points. There are similar, but smaller, drops in membership for those who moved to quartile 2 (yellow) or quartile 3 (red) districts. However, membership rates are much more stable for those who moved to other quartile 4 districts (blue).

Likewise, there is symmetry when examining those teachers who moved away from first quartile districts (right panel). For instance, the average membership rate increases from about 10 percent to over 30 percent for those teachers who moved to a fourth-quartile district (blue).

This event study analysis suggests a complementary relationship, in which the district membership rate plays a substantial role in one's own membership decision. However, this movers design is only for people who actually switched to a new workplace by definition. I therefore cannot disentangle whether this is because of social factors in the new workplace, or because of other factors such as the quality of the union in a particular district. This motivates the experimental design

[5] See Card, Heining, and Kline (2013) for a similar analysis on workers moving between high- and low-paying firms.

where I shift teachers' beliefs about their district's membership rate, without requiring a move to a new workplace.

B. Hypotheses Tested

The information provision experiment has four main objectives as outlined in the pre-analysis plan. First, I gauge to what degree teachers misperceive their district's membership rate. Second, I estimate whether treated subjects update their beliefs relative to the control group after receiving a truthful estimate of their district's rate. Third, I examine whether subjects who update their beliefs then change either their planned or actual union participation in the following school year. Importantly, I separately measure effects for people who overestimated and underestimated the district membership rate. This will allow me to see whether individual membership behavior is related to the group membership rate either as strategic complements (as suggested by the movers analysis) or strategic substitutes (as in the literature on protest participation). Finally, I assess treatment effect heterogeneity, especially by baseline union membership at the individual- and district-level.

C. Experimental Design

In the spring of 2024, I sent a Qualtrics survey via email to almost 45,000 teachers in the state of Wisconsin. The recruitment sample included almost all public-school teachers in the state with a few exceptions. First, I could not find publicly available staff emails for roughly 13 percent of the school districts. Second, I did not send the experimental survey to school districts that were not affiliated with the NEA or to districts for which my estimated union membership rate was below 5 percent, to ensure at least some union presence.

The email invited teachers to participate in a 5–10 minute survey about their perceptions of their jobs and unions.[6] As compensation, respondents were entered into a raffle drawing where 60 winners received $50 Amazon gift cards. I sent the first batch of email invitations on April 10 and proceeded in a staggered fashion until April 30.[7] The survey was closed and the raffle drawing was conducted on May 14.

In the survey, I asked all respondents to estimate what share of teachers in their school district were members of the union last year. This response, which importantly was asked of both treatment and control groups, serves as the primary measure of prior beliefs.[8] After eliciting prior beliefs,

[6]Some of the pre-experimental responses were used in Foy (2025) to understand differences in teacher perceptions across union regimes.

[7]I take advantage of the staggered rollout in the robustness exercises in Section V to examine potential spillover effects.

[8]Additionally, I asked respondents to estimate the share of members in their specific school. I discuss this further when assessing robustness in Section V.

I then provided the information provision. If randomized to the treatment group, participants saw the following text:

*Our best estimate is that X% of teachers in your school district were members of the teachers' union last year.**

In place of the "X", respondents saw an estimate of their district-specific membership rate. As seen in Appendix Figure A1, there is considerable variation in the district membership rates ranging from below 10 percent to above 80 percent.[9]

Next, I elicited respondents' posterior beliefs from both treatment and control groups by asking them to predict their district's membership rate in the next school year. By asking about next year's membership rate, I captured updated expectations without repeating the same question twice.

Finally, I asked all respondents how likely they were to be a member of the teachers' union next year on a five-point scale ranging from very unlikely to very likely. This question serves as the basis for the outcome variable on future planned participation.

The survey also asked questions measuring respondents' baseline pro-sociality. Following Enke, Rodríguez-Padilla, and Zimmermann (2023), I presented hypothetical scenarios where respondents were asked how they would split a 10 percent budget surplus with various groups. In three separate scenarios, respondents are asked what percentage of the surplus they would keep for teacher pay versus using it for support staff pay, student program funding, and administrator pay. The administrator pay category was included to confirm that teachers were paying attention with the assumption that fewer teachers would split the surplus with their superiors. Appendix Figure A2 panel A confirms this, as the average share given to support staff, students, and administrators was 43 percent, 40 percent, and 17 percent, respectively. I classify teachers as "pro-social" if they split the surplus at least 50-50 with either one or two of the three groups.[10]

The survey ended with an attention check following Bottan and Perez-Truglia (2025). About 98 percent of teachers passed the attention check. As such, I include all respondents in the primary analysis sample, but results are robust to excluding the 2 percent who answered incorrectly.

The experiment has two primary outcome measures. The first is a survey-based outcome I call "planned participation," which is whether the respondent stated that they intended to be in the union in the following school year. I denote someone as a planned member if they said they were "very likely" or "somewhat likely" to be in the union next year. The second is an administrative data outcome on whether the respondent was actually a member of the teachers' union in the

[9] The asterisk pointed to a link where respondents could see more information on how the estimate was constructed. See Appendix Section C for an example of the experimental survey.

[10] Using this measure, I find that union members are more pro-social on average (Appendix Figure A2 panel B).

following school year (2024–25), again using the campaign finance reports. This allows me to examine whether the information provision affected one's union membership status between the 2023–24 and 2024–25 school years.

IV. Empirical Strategy and Sample Characteristics

A. Empirical Strategy

The primary research question is whether teachers update their beliefs in response to the information treatment and whether they subsequently alter their planned or actual union status. To answer this, I estimate a two-stage least squares regression, where the treatment serves as an instrument for the change in posterior beliefs. Formally, the first stage equation is

$$D_i = \alpha + \pi Z_i + X_i'\eta + u_i, \tag{1}$$

where D_i is a teacher's posterior belief about the union membership rate in their school district and Z_i is the treatment indicator. Following Cantoni et al. (2019), I code the treatment variable as "-1" for those respondents who overestimated their district rate to measure how treatment moved people towards the truthful estimate. The vector X_i' includes controls for prior beliefs, individual membership status prior to the survey, and additional baseline controls to improve precision.

The reduced form equation that relates the outcome variables to treatment is

$$Y_i = \alpha + \rho Z_i + X_i'\eta + u_i, \tag{2}$$

where the terms are as before but now the outcome (Y_i) is an indicator variable for planned or actual union participation. Therefore, the second-stage equation,

$$Y_i = \alpha + \beta D_i + X_i'\eta + u_i, \tag{3}$$

measures how much membership rates (Y_i) change due to a corresponding change in beliefs (D_i). It should be noted that the exclusion restriction may not strictly hold because, in addition to changing beliefs, the treatment may change other perceptions such as beliefs about the quality of the union. I discuss this further in Section VI.

The primary analysis pools respondents who underestimate and overestimate their district-specific membership rate. As such, I also include an indicator variable for overestimating the district rate and its interaction with each control variable. This is necessary to include because overestimators and underestimators may react differently to treatment. To understand this further, I also report results for underestimators and overestimators separately.

B. Sample Characteristics

I first examine observable characteristics of the teachers who participated in the experiment relative to the broader teaching workforce. I then check whether the treatment and control groups are balanced on these characteristics.

Participation in the Study: The experimental survey was sent to 43,637 teachers across the state of Wisconsin. Of those invitations, 2,789 individuals filled out at least one question of the survey and 2,525 completed enough of the survey such that they were randomized to one of the two treatment arms. This implies a response rate of about 5.8 percent, which is similar to other studies that conduct cold recruitment strategies.[11] I further drop 33 respondents who were either not licensed teachers or could not be matched to the administrative personnel data (see Appendix Section B for more details). After these restrictions, there are 2,492 individuals in the experimental study, 1,232 randomized to the treatment group and 1,260 to the control group. In the main analysis, I also exclude 70 respondents who did not answer any of the three questions on prior beliefs, posterior beliefs, or planned participation.

A benefit of the linked data is that I can observe baseline characteristics for both survey respondents and those who received the survey but did not participate. Table 1 compares the survey participants to non-respondents within the same district. Column 1 shows the baseline mean characteristics for non-participants and column 2 shows the difference for those in the experimental study. In general, there are statistically significant though small differences: for instance, respondents had slightly more experience, were more likely to be working full time or in multiple roles, and were less likely to be math teachers. The survey sample was also more likely to be male and less likely to be non-white. Of particular interest, the survey participants were 6 percentage points more likely to be union members via the matching procedure at baseline, a difference of roughly 15 percent. This selection by membership status is consistent with the recruitment email stating that the survey was on teachers' perceptions of unions. Conveniently, the survey sample had a roughly equal split of members and non-members, which allows me to examine differences by baseline union status.

Balance Across Treatment Groups: More importantly for the validity of the design, I examine whether the treatment and control groups were balanced across observable characteristics in Table 2. In addition to the baseline characteristics from Table 1, I test for differences in baseline

[11] It was also infeasible to provide payments to everyone who participated given the large potential sample size. Most similarly, Giaccobasso et al. (2025) sent invitations to residents in Dallas County via mail and received a response rate of 3.6 percent. Cantoni et al. (2019) received a response rate of 19.1 percent using a sample of undergraduates from Hong Kong University of Science and Technology, though all participants were paid.

pro-sociality and prior beliefs about their district and school membership rates. Column 1 shows the average rate of each characteristic in the control group and column 2 shows the difference in the treatment group. The observable characteristics are well balanced across treatment arms, indicating that the randomization was successful. Across the fourteen variables, there are only two that are statistically significant at the 10 percent level: treated respondents were slightly less likely to be male (p-value = 0.06) and slightly less likely to be non-white (p-value = 0.06). Moreover, when regressing the treatment indicator on the full set of covariates, I fail to reject that baseline characteristics are jointly the same (F-test p-value = 0.53).

I also examine balance separately for underestimators and overestimators in Appendix Tables A1–A2, since the main analysis pools these two groups. Once again, covariates are well-balanced across treatment groups. Though not statistically significant, the point estimate on baseline membership for underestimators was -3.2 percentage points (p-value = 0.29), indicating that treated respondents were somewhat less likely to be union members. This is an important control in the analysis given that baseline membership very strongly predicts future planned or actual membership status.

Prior Beliefs: Figure 2 plots the distribution of prior beliefs relative to the district membership rate, where the district rate is normalized to zero. The figure shows that the modal person had accurate perceptions of membership in their district.[12] However, there is considerable variation to the left and right of zero, implying that many people both underestimated and overestimated their district rate, respectively. There was slightly greater overestimation: the average estimate was about 7 percentage points above the district rate.

Certain teachers are more accurate than others. For instance, Appendix Table A3 column 1 shows that baseline union members had priors that were 1.8 percentage points closer to the truth relative to non-members, a difference of about 11 percent.[13] Likewise, teachers with above-median experience in the district had priors that were 2.4 percentage points closer to the truth compared to less-experienced teachers. I also asked respondents whether they knew their local union's legal status, as the State of Wisconsin requires that unions hold annual elections to recertify their right to collectively bargain (Foy 2025). Using the fall 2023 certification election results, I find that, again, members and more experienced teachers were significantly more likely to correctly identify the certification status in their district (Table A3 column 3). For example, members were 10 percentage points more likely to correctly identify the certification status, a difference of about 13 percent.

[12]The centering around 0 is not due to people guessing 50 percent when they are uncertain (Fischhoff and De Bruin 1999). Appendix Figure A3 shows that this pattern holds even when restricting to districts with a membership rate below 40 percent or above 60 percent.

[13]Members were also more likely to overestimate the district rate compared to non-members (Appendix Table A3 column 2), suggesting that they had more optimistic estimates than people who were not part of the union.

Together, these results strongly imply that members and experienced teachers have a better sense of the quality of the union at baseline, as they had more informed priors about the district membership rate and certification status.

V. Experimental Results

A. First Stage Effect on Beliefs

I first examine whether the experimental intervention caused respondents in the treatment group to update their beliefs about the union membership rate in their district. Table 3 presents the results of equation 1, estimating whether the intervention affected posterior beliefs. In panel A, I show results for the pooled sample where treatment is recoded as -1 for those who overestimate their district rate. Column 1 includes no controls except for one's prior belief.[14] Column 2 adds a control for whether the teacher was a union member at the start of the school year. I include additional baseline controls in column 3 to improve precision.[15] Across specifications, there is a persistent, statistically significant coefficient of about 8.5 percentage points (p-value < 0.001), which indicates that treated respondents were more likely to update their beliefs towards the provided estimate. This is a substantial reduction in misperceptions as the baseline absolute gap between prior beliefs and the district rate was about 17 percentage points.

Panels B and C of Table 3 report results separately for underestimators and overestimators, respectively. Underestimators increased their posterior beliefs by about 5 percentage points, while overestimators reduced their beliefs by about 11 percentage points. This represents substantial updating for both groups, given that baseline misperceptions were 12 and 21 percentage points for the two groups, respectively.

To examine the updating visually, I plot the distribution of priors (gray) overlaid with the distribution of posteriors (blue) in Figure 3. Panel A illustrates that the control group's posterior beliefs are very similar to their prior beliefs. Panel B shows the corresponding figure for the treatment group. Here, the posterior beliefs are much more concentrated to the center of the figure, indicating that many treated respondents updated exactly to the provided estimate.

To more formally estimate the degree of updating in the treatment group, I calculate learning rates following Cavallo, Cruces, and Perez-Truglia (2017). This procedure estimates the weight that treated participants put on the signal versus the weight they placed on their prior belief. The

[14] Additionally, the pooled sample includes a control for whether the respondent's prior belief was above the district membership share as well as an interaction between the indicator and prior beliefs.

[15] The additional controls include years of experience in the district, full-time employment percentage, number of positions, whether the teacher had an advanced degree, whether they taught reading or math, whether the teacher was male or non-white, and fixed effects for each of the 13 union regions. See Appendix Section B for more details.

estimating equation is

$$\text{Updating}_i = \alpha + \beta_1 Z_i \times \text{Perception gap}_i + \beta_2 Z_i + \beta_3 \text{Perception gap}_i + \epsilon_i, \tag{4}$$

where Updating_i is the difference between an individual's posterior and their prior belief and Perception gap_i is the difference between the district rate and an individual's prior belief. The coefficient of interest, β_1, represents the degree of updating that occurs for members of the treatment group net of any updating in the control group. A coefficient of 1 would indicate that teachers in the treatment group perfectly update their beliefs towards the provided estimate. As seen in Appendix Table A4, the β_1 coefficient is 0.55, which means that teachers in the treatment group placed a weight of 55 percent on the provided signal and a weight of 45 percent on their own prior belief. A learning rate of 0.55 is consistent with estimates from other information provision experiments (see Haaland, Roth, and Wohlfart (2023) for a review).

To summarize, the first stage analysis shows that the information provision worked as intended — individuals in the treatment group substantially reduced their misperceptions relative to those in the control group. Moreover, the reduction in misperceptions was roughly equal for underestimators and overestimators. Given the strong first stage, I next turn to whether updating one's beliefs about the district membership rate affected intentions.

B. Effects on Planned Union Participation

Reduced Form Effect: I next examine whether the first-stage belief updating translated into changes in planned participation. I present the reduced-form results from Equation 2 in Table 4, where once again column 1 shows results without any controls, column 2 adds a control for an individual's baseline membership status, and column 3 includes additional controls to improve precision. For simplicity, I recode the likelihood of participation question as an indicator variable equal to 100 if respondents said they are *very likely* or *somewhat likely* to be members of the teachers' union next year and 0 if they responded *uncertain*, *somewhat unlikely*, or *very unlikely*. Focusing on the pooled sample in Panel A, I find a positive but insignificant coefficient of 2.5 percentage points in column 1. This points to complementarity in membership but the estimate is imprecise. Recall, however, that there was slight imbalance in treatment assignment between members and non-members for underestimators, which is important as prior membership is very predictive of future membership behavior. When controlling for just prior membership in column 2, I find a similar but more precise coefficient of 2.7 percentage points (p-value = 0.05). This indicates that planned participation is complementary to the group membership rate. In column 3, there is a very similar estimate of 3.2 percentage points when adding additional controls (p-value = 0.02).

As in Table 3, panels B and C of Table 4 split the sample by underestimators and overestimators, respectively. My preferred specification with controls in column 3 shows that there is about a 2–4 percentage point effect in the direction of the provided estimate for both underestimators and overestimators. This indicates that the pooled effect is not driven by either group; in both cases, planned participation moves in the same direction as the updated first-stage beliefs.

The prior analysis defined planned participation as an indicator variable for if a respondent said they were "very likely" or "somewhat likely" to be a member of the union next year. However, this masks some of the variation since the question was on a 5-point scale, especially since 12 percent of the sample said that they were "undecided." In Appendix Table A5, I instead use a scale from 1 (very unlikely) to 5 (very likely) as the outcome variable. This analysis shows relatively more precise estimates in favor of complementarity.

2SLS Estimates on Planned Participation: Tables 3 and 4 showed evidence of strategic complementarity. Treated respondents who underestimated the district rate were more likely to say they would be a member next year, while those who overestimated the district rate were less likely to say they would be a member next year. In Table 5, I put these pieces together in a two-stage least squares regression, where the change in beliefs is instrumented by treatment assignment. Focusing on the results with controls in column 3, the coefficient of 0.36 indicates that a 10 percentage point increase in beliefs about the district membership rate increases the likelihood that one plans to be a member by 3.6 percentage points. This relies on the assumption that treatment affects planned participation only through updating on the group membership rate. This may not be strictly true because people may also update on other items, such as the quality of the local union. I discuss this further in Section VI.

Magnitude: The 2SLS estimates can be viewed in the context of a behavioral elasticity. For instance, an elasticity of 1 would imply that an individual's membership decision corresponds one-to-one with their beliefs about others' participation. Conversely, an elasticity of 0 would imply that individuals are not affected at all by others' participation. The 2SLS estimates indicate that a 10 percentage point increase in beliefs about others' participation induces a 3.6 percentage point increase in one's own planned participation. This point estimate implies a behavioral elasticity of roughly 0.36 because the baseline rate of actual membership (0.44) is roughly equivalent to the baseline belief of others' membership (0.45). While significantly less than one, this estimate represents a substantial correspondence between others' participation and one's own intentions.

As a non-experimental benchmark, I regress individual membership status on the annual change in district membership rates using the full administrative data. This exercise mirrors the event study analysis in Section III. Specifically, for each teacher in the administrative data, I create leave-one-out district-level membership rates in each year. I then estimate the change in a person's district membership rate from the prior year to the current year. As previously discussed, this non-experimental analysis is not necessarily causal, because other factors could affect both the change in district rates and one's own membership decision such as a change to the quality of the union or administration. Nevertheless, I show the results from this analysis in Appendix Table A6 because the estimates are very similar to the experimental elasticity. In column 1, I find that a 1 percentage point change in district-level membership rates corresponds to a 0.37 percentage point change in one's own membership status. Results are similar when adding controls in column 2.

Alternative Specifications: I examine robustness of the main results on planned participation in Appendix Table A7. Panels A–C present results for the first stage, reduced form, and two-stage least squares specifications, respectively. For reference, column 1 shows the baseline result with controls from column 3 of Tables 3–5. In column 2, I incorporate the fact that the survey asked teachers to estimate the union membership rates in their school in addition to their district. Not surprisingly, district and school beliefs are highly correlated. The column 2 estimates replace the prior and posterior district predictions with the corresponding school variables. Results are similar, suggesting that the treatment induced people to update not only on their district but on the union membership rates in their actual school building.

Although treatment is assigned at the individual level, there is still likely serial correlation in union membership within districts. Column 3 addresses this by including district fixed effects and clustering standard errors at the district level. There are slightly fewer observations, because some districts do not have both treatment and control observations, but the results are relatively more pronounced and precise than in column 1.

One issue with the pooled sample is that it is not clear how to assign teachers in the treatment group who estimate the district rate exactly. Column 4 presents an alternative model, in which these treated respondents are re-coded as 0 and included in the control group. The logic is that these people should not change their behavior given that they are already correctly estimating the district rate. This change has little impact on the results.

Finally, I impose two sample restrictions in columns 5–6. In column 5, I drop the respondents who failed the attention check at the end of the survey. In column 6, I drop outliers by excluding 1 percent of the sample with the most extreme prior beliefs. Again, results are similar in these two subsamples.

An additional concern is that the self-reported findings are influenced by experimenter demand bias, where participants alter their stated behavior because of researcher expectations. While I cannot completely rule this out, it is reassuring that I find similar effects for both underestimators and overestimators. Underestimators may feel pressure to say they will be members upon learning that the district estimate is higher than they thought. However, the effect of demand bias on overestimators is somewhat less clear — they may not perceive a lower estimate as a researcher endorsement.

Spillover Effects: A related concern is that treated respondents informed others, particularly control group members, about the average membership rate. Since the survey went out over email, it is possible that teachers in a school took it at the same time or discussed it with colleagues. If anything, this would likely bias results towards zero given that this would result in the control group having more informed priors.

Nevertheless, several pieces of evidence suggest limited spillovers. For one, most schools had few participants as the 2,422 respondents in the main sample were from 976 unique schools. Additionally, control group participants had posteriors that were very similar to their priors (Figure 3 panel A). To test this more formally, I exploit the fact that the survey was administered over a five-week time frame by testing whether later respondents had more accurate prior beliefs relative to those who took the survey earlier. In Appendix Figure A4, I regress the gap between prior beliefs and the district share on survey week fixed effects. The results show that the gap in prior beliefs was not significantly different across weeks. This suggests that the information provision was not being shared across teachers.

The preceding analysis showed that a change in beliefs caused respondents to change their union intentions in a complementary fashion. Given these strong results on planned participation, I next turn to whether this translated into changes in actual union participation the following school year.

C. Effects on Actual Union Participation in the Following School Year

I now examine whether the short-run effects on planned participation translated to changes in respondents' actual membership decisions in the following school year. Since the experiment ran in spring 2024, any membership changes would likely occur in the summer when teachers are considering dues payment for the upcoming school year. To measure actual membership status, I again match the personnel data with the campaign finance records to create an indicator for membership status in the 2024–2025 school year.

I estimate the same reduced form and 2SLS regressions as in Tables 4–5, but the samples are slightly smaller for two reasons. First, I exclude 138 people who are not in the Wisconsin public-school staff data file in 2024–2025. These people have left the Wisconsin teaching workforce and therefore I cannot observe whether they are union members in the following year. I also exclude 32 people who are in the personnel file but not working as teachers. These people have moved to other roles, such as substitute teachers or principals, and therefore do not pay dues to the state teachers' union.[16]

Reduced Form Effect on Actual Participation: In Table 6, I present the reduced form results with 2024–25 membership status (multiplied by 100) as the outcome variable. As with Table 4, I show results with no controls (column 1), controlling for baseline membership status (column 2), and with additional controls (column 3). Panel A displays results for the pooled sample and panels B–C show results for underestimators and overestimators, respectively. In panel A column 3, I find a positive coefficient of 0.93 that is not statistically different from zero. This indicates that the sign of the relationship is consistent with the results on stated intentions, but the estimate is closer to 0 and less precise. In panels B–C, the coefficients are imprecise but the sign of the estimates point to complementarity rather than substitutability.

2SLS Estimates on Actual Participation: Table 7 shows the results on actual union participation in the 2SLS framework. Again, the results are of the same sign as for stated intentions, but insignificant and closer to zero. Of note, the underestimators coefficient in panel B is 0.30, which is imprecise but broadly similar to the results on stated intentions. Conversely, the coefficient for overestimators is substantially smaller at 0.03. This may suggest a larger importance of complementarity for those who underestimated their district's membership rate, but it is hard to say definitively given the imprecision.

To summarize, I find that the experimental treatment induced a change in beliefs towards the provided district membership rate. Accordingly, treated respondents' planned participation changed in the same direction as their change in beliefs. However, that change in intentions did not fully translate into respondents changing their actual membership status. I next examine whether effects were more or less pronounced for various subgroups.

VI. Mechanisms

In this section, I explore potential mechanisms by examining heterogeneity on three pre-specified dimensions: the baseline membership rate in the district, whether the individual was a union member at baseline, and whether the respondent is pro-social as measured by the survey vignette. While not pre-specified, I also examine differences by respondent's experience level in the district, given the strong correlation between experience and correct priors about the union (Appendix Table A3). The heterogeneity analysis helps to distinguish between two mechanisms. The first is a

[16] There are also three respondents included in the actual participation analysis who are excluded in the planned participation analysis because they skipped the planned participation question. Table A8 shows that the 2SLS results on planned participation are robust, and relatively larger, when excluding the respondents who leave the teaching workforce.

social mechanism whereby individuals respond directly to the share of their colleagues who are members. In this case, we would expect results to be stronger in districts with more members at baseline, given that the penalty for free-riding is likely higher when more people participate. The second mechanism is that respondents update on the quality of their local union. If the effects are driven by updated views on union quality, then we would expect non-members and inexperienced teachers to drive the results, given that these teachers are likely less informed about union quality relative to members and experienced teachers at baseline.

To examine this, I separately estimate the 2SLS results by various subgroups. Figure 4 plots 2SLS coefficients for each dimension of heterogeneity. The results on stated intentions are in panel A and the results on actual union participation in the following year are in panel B. Overall, the estimates for members and experienced teachers tend to be larger, suggesting that union quality is not the primary mechanism. On the other hand, I find suggestive evidence that the results are most pronounced for districts where a majority of people are members. This aligns with the idea that the social customs around membership are eroded when less than half of teachers are in the union.

District Membership Rate: I first split the sample by whether the survey respondent was in a district with either above 50 percent membership or below 50 percent membership. This is an interesting threshold for two reasons: first, social norms, such as peer pressure, are likely more relevant in districts where a majority of people are members. Conversely, a teacher may not be persuaded by the information provision if she finds out that few people are in the union, regardless of whether she overestimated or underestimated the district rate. Second, Wisconsin teachers' unions need to hold annual elections to recertify their legal standing in the state. This requirement likely makes the 50 percent threshold very salient for teachers, especially in districts that have maintained their certification status.[17]

For planned participation (panel A), I find similar effects for under- and over-50 percent districts, though the over-50 percent group has a larger 2SLS coefficient (0.48 compared to 0.30). For actual participation, I find suggestive evidence that the planned participation result carried through to actual participation. Specifically, the coefficient for actual participation in the above-50 percent group was 0.52 (p-value = 0.03). For under-50 percent districts, I find an effect centered around 0 and insignificant. One concern with this result is that respondents in the over-50 percent districts might be more likely to be underestimators mechanically (and vice versa), since there is less room to overestimate in a district with a high membership share. However, even in above-50 percent districts, 60 percent of respondents overestimate the district rate (65 percent do so in below-50

[17]Technically, the threshold for recertification is 51 percent of employees voting in favor of recertification. I present results at the 50 percent threshold for simplicity, but results are similar when instead using the 51 percent threshold. Note also that the share of teachers voting for recertification is higher than those in the union, as teachers can be in favor of recertification, but not actually a dues-paying member (see Foy 2025).

percent districts), implying that people tend to be overoptimistic across districts. To investigate this further, Appendix Tables A9–A10 show the full 2SLS results for above-50 percent and below-50 percent districts, respectively. For above-50 percent districts, I find large point estimates for both underestimators and overestimators, but null results in both directions for below-50 percent districts. This indicates that the complementarity between the individual and group membership rates was especially pronounced in districts where a majority of teachers were members.[18]

Individual Teacher Characteristics: Next, I split the sample by individual respondent characteristics, such as baseline membership status. This is an important source of heterogeneity because it is possible that the information provision shifted not only beliefs about others' participation but also about the quality of the union. For instance, perhaps someone who underestimated the district rate now has a more favorable view of the union's bargaining power. Presumably, people who were in the union at baseline have a good idea about the quality of the union already. In support of this assumption, I find that members have more accurate priors about the district membership rate and the district's bargaining status (Appendix Table A3). For stated intentions in panel A, I find similar 2SLS coefficients for non-members (0.36) and members (0.31). For actual participation in panel B, I estimate a 0.04 coefficient for non-members and a 0.20 coefficient for members. Overall, this provides suggestive evidence that people are responding to their updated beliefs about the share of members and not on union quality since baseline members respond at least as strongly as non-members.

I also split the sample by whether a respondent had below- or above-median experience within the district. As with membership status, high-experience teachers likely have more knowledge about union quality given that they have more informed priors (Appendix Table A3). Again, I find suggestive evidence that more experienced teachers have stronger responses than less experienced teachers. For high-experience teachers, the 2SLS coefficient on actual participation is 0.25 (p-value = 0.05). For low-experience teachers, the 2SLS coefficient is near zero and insignificant. Together, the split-sample results are not consistent with respondents updating on the quality of the union, given that members and experienced teachers were likely more informed about the union before the experiment.

Lastly, I examine whether results differ by the pro-sociality measure that was elicited in the survey. Recall that I asked each person how they would split a 10 percent budget surplus between teacher pay and either support staff pay or student program funding. The pro-social measure is equal to 1 if the respondent split at least half the surplus with either group and 2 if they split at least

[18]The pre-analysis plan specified looking at effects by 10 percentage point bins of the district membership rate. However, there is limited ability to detect effects given the sample sizes within each bin. For transparency, I present these results in Appendix Figure A5.

half with both groups. On the one hand, prosocial individuals may be less likely to change their behavior in response to the treatment because their actions are motivated by altruism, rather than by others' participation. Conversely, prosocial individuals may be more responsive to changes in group membership rates precisely because they are more attuned to social norms and the behavior of others.

I examine heterogeneity by pro-sociality in the bottom row of Figure 4. In panel A, there is some evidence of stronger results for the most pro-social people according to the elicitation vignette. However, this pattern did not carry forward to actual membership status (panel B), making it hard to draw definitive conclusions.

VII. Conclusion

This study investigates whether teachers update their beliefs, intentions, and actual union membership status following an exogenous shift in their perceptions of others' participation in the union. To examine this question, I conduct an information provision experiment in which treated respondents receive a truthful estimate of their school district's union membership rate. I find that the information provision closes about half of the 17 percentage point gap between teachers' prior beliefs and the district estimate. Accordingly, teachers in the treatment group were more likely to adjust their plans for union participation in the direction of their revised beliefs. I estimate that a 10 percentage point change in beliefs leads to a 3.6 percentage point shift in planned participation. This indicates that individuals' reported membership decisions are complementary to their beliefs about others' participation. Notably, this effect is in the opposite direction of other recent experimental papers studying collective action problems that emerge during protests (Cantoni et al. 2019; Jarke-Neuert, Perino, and Schwickert 2023) and election campaigning (Hager et al. 2023). One possible explanation for the opposite result is that social considerations are more important in the context of union membership within a workplace compared to large settings where free-riding is less costly.

In terms of actual union participation, I estimate a positive coefficient in line with complementarity but cannot reject that the effect is equal to 0. This implies that the intervention shifted stated intentions, but was not enough to meaningfully change actual participation over a longer horizon. One conclusion is that it therefore takes more than a simple informational push to change actual union participation decisions. Yet, I find suggestive evidence of a larger response for respondents in districts where a majority of teachers are union members. This may indicate that social customs, such as peer pressure, are relevant but only when a majority of individuals within an organization participate in the union.

References

Akerlof, George A. 1980. "A Theory of Social Custom, of Which Unemployment May be One Consequence." *Quarterly Journal of Economics* 94: 749–75.

Booth, Alison L. 1985. "The Free Rider Problem and a Social Custom Model of Trade Union Membership." *Quarterly Journal of Economics* 100(1): 253–61.

Bottan, Nicolas, and Ricardo Perez-Truglia. 2025. "Betting on the House: Subjective Expectations and Market Choices." *American Economic Journal: Applied Economics* 17(1): 459–500.

Cantoni, Davide, David Y. Yang, Noam Yuchtman, and Y. Jane Zhang. 2019. "Protests as Strategic Games: Experimental Evidence from Hong Kong's Anti-Authoritarian Movement." *Quarterly Journal of Economics* 134(2): 1021–77.

Card, David, Jörg Heining, and Patrick Kline. 2013. "Workplace Heterogeneity and the Rise of West German Wage Inequality." *Quarterly Journal of Economics* 128(3): 967–1015.

Cavallo, Alberto, Guillermo Cruces, and Ricardo Perez-Truglia. 2017. "Inflation Expectations, Learning, and Supermarket Prices: Evidence from Survey Experiments." *American Economic Journal: Macroeconomics* 9(3): 1–35.

Ellwood, David T., and Glenn Fine. 1987. "The Impact of Right-to-Work Laws on Union Organizing." *Journal of Political Economy* 95(2): 250–73.

Enke, Benjamin, Ricardo Rodríguez-Padilla, and Florian Zimmermann. 2023. "Moral Universalism and the Structure of Ideology." *Review of Economic Studies* 90: 1934–62.

Fischbacher, Urs, Simon Gächter, and Ernst Fehr. 2001. "Are People Conditionally Cooperative? Evidence from a Public Goods Experiment." *Economics Letters* 71(3): 397–404.

Fischhoff, Baruch, and Wandi Bruine de Bruin. 1999. "Fifty-Fifty=50%?" *Journal of Behavioral Decision Making* 12: 149–63.

Foy, Morgan. 2025. "Selection and Performance in Teachers' Unions." Working Paper.

Giaccobasso, Matias, Brad Nathan, Ricardo Perez-Truglia, and Alejandro Zentner. 2025. "Where Do My Tax Dollars Go? Tax Morale Effects of Perceived Government Spending." *American Economic Journal: Applied Economics* 17(4): 223–59.

González, Felipe. 2020. "Collective Action in Networks: Evidence from the Chilean Student Movement." *Journal of Public Economics* 188: 104220.

Haaland, Ingar, Christopher Roth, and Johannes Wohlfart. 2023. "Designing Information Provision Experiments." *Journal of Economic Literature* 61(1): 3–40.

Hager, Anselm, Lukas Hensel, Johannes Hermle, and Christopher Roth. 2022. "Group Size and Protest Mobilization across Movements and Countermovements." *American Political Science Review* 116(3): 1051–66.

Hager, Anselm, Lukas Hensel, Johannes Hermle, and Christopher Roth. 2023. “Political Activists as Free Riders: Evidence from a Natural Field Experiment.” *Economic Journal* 133: 2068–84.

Hertel-Fernandez, Alexander, Suresh Naidu, Adam Reich, and Patrick Youngblood. 2020. “Understanding the COVID-19 Workplace: Evidence from a Survey of Essential Workers.” New York: Roosevelt Institute.

Janus v. AFSCME. 2018. No. 16–1466. https://www.supremecourt.gov/opinions/17pdf/16-1466_2b3j.pdf.

Jarke-Neuert, Johannes, Grischa Perino, and Henrike Schwickert. 2023. “Free Riding in Climate Protests.” *Nature Climate Change* 13: 1197–1202.

Lumsden, Keith G., and Craig H. Petersen. 1975. “The Effect of Right-to-Work Laws on Unionization in the United States.” *Journal of Political Economy* 83(6): 1237–48.

Manski, Charles F. 1993. “Identification of Endogenous Social Effects: The Reflection Problem.” *The Review of Economic Studies* 60: 531–42.

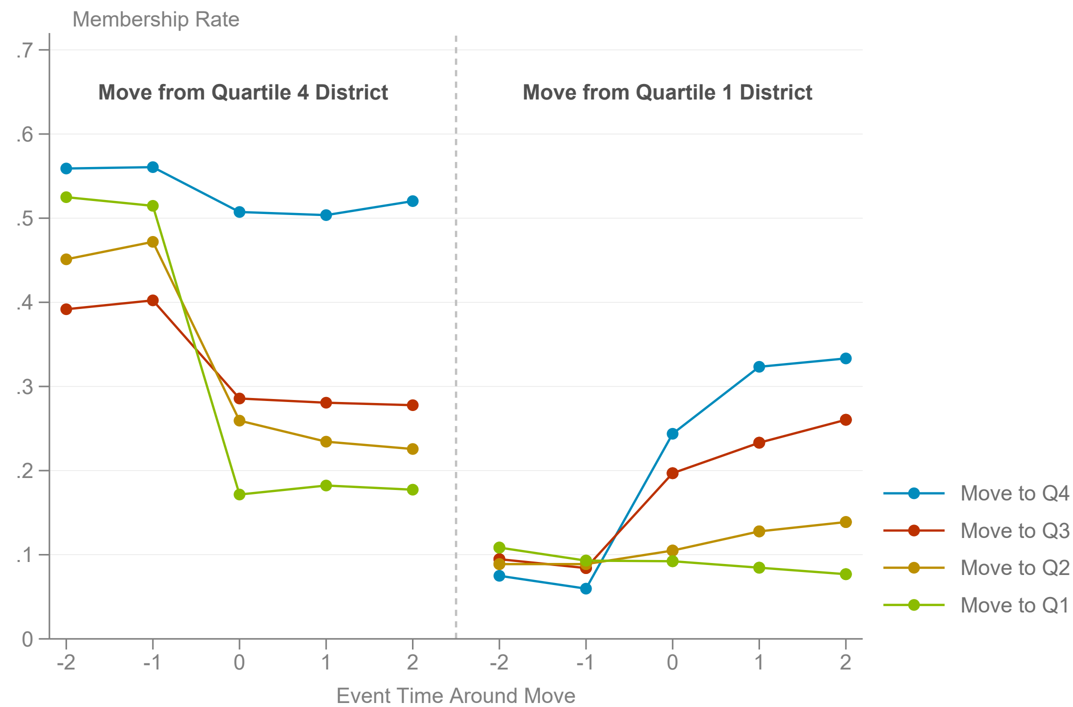
Naidu, Suresh. 2022. “Is There Any Future for a US Labor Movement?” *Journal of Economic Perspectives* 36(4): 3–28.

National Center for Education Statistics. 2021. “National Teacher and Principal Survey, 2021.” Accessed December 2023. https://nces.ed.gov/surveys/ntps/.

Naylor, Robin, and Martin Cripps. 1993. “An Economic Theory of the Open Shop Trade Union.” *European Economic Review* 37: 1599–620.

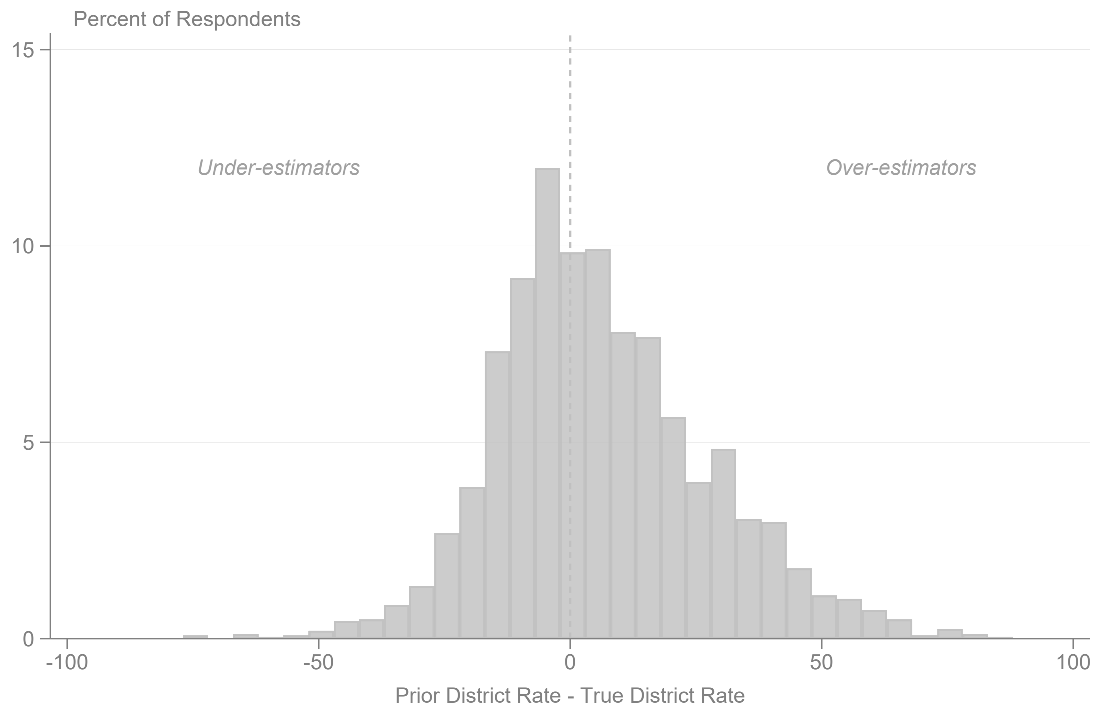
Olson, Mancur. 1965. *The Logic of Collective Action: Public Goods and the Theory of Groups*. Cambridge: Harvard University Press.

Figure 1: Event Study of Membership Rates around Moves to New Districts



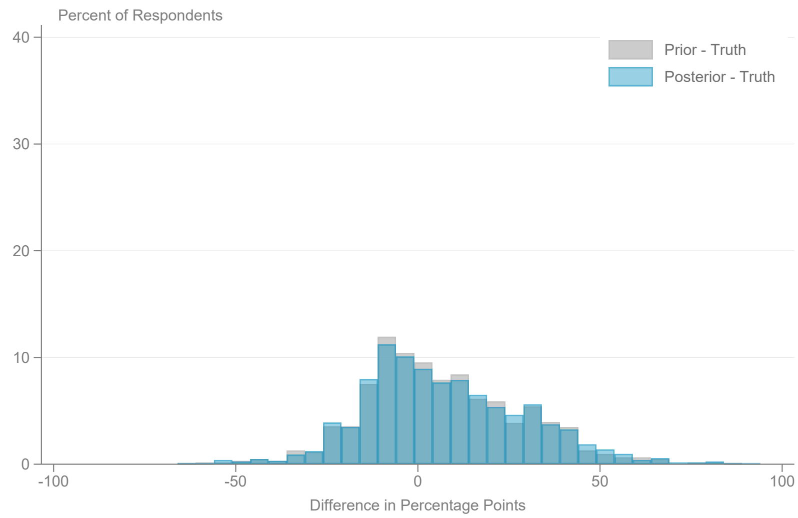
Note: The figure shows individual membership rates around the timing of a move to a new school district. The sample is teachers who moved once between 2018–2020 and have two years of data pre- and post-move. I split the sample into four quartiles of 2017 membership rates (using all teachers not just movers). I then separately examine membership rates for people who moved from a district in each of the four quartiles to a new district in any of the four quartiles. For simplicity, the figure shows only moves from quartile four (left panel) and quartile one (right panel). Individuals who moved to a new district in quartile four, three, two, or one are shown in blue, red, yellow, and green, respectively.

Figure 2: Distribution of Priors around District Estimate of Membership Rate

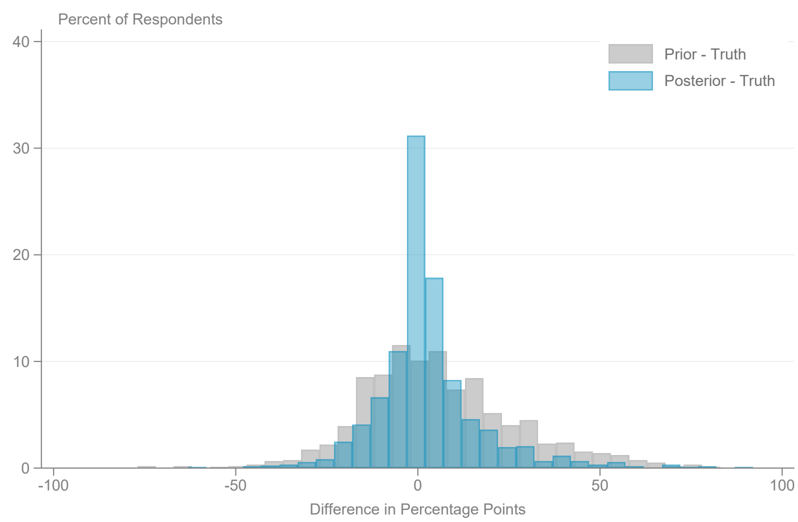


Note: The x-axis is an individual's prior belief of their school district's membership rate minus the truthful estimate of the district membership rate. The district rate is normalized to 0. The y-axis is the share of respondents who fall into each 5 percentage point bin. Respondents who underestimated their district share lie to the left of the dashed line; respondents who overestimated their district share lie to the right of the dashed line.

Figure 3: Belief Updating: Distribution of Priors and Posteriors



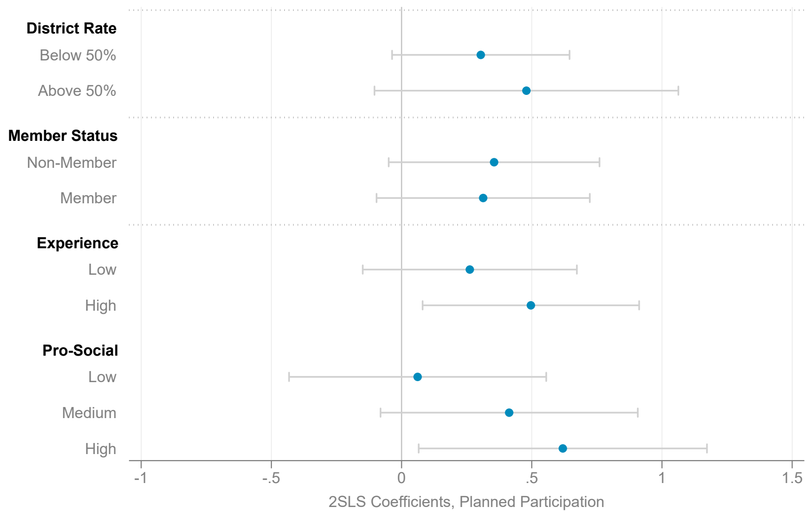
(a) Control Group



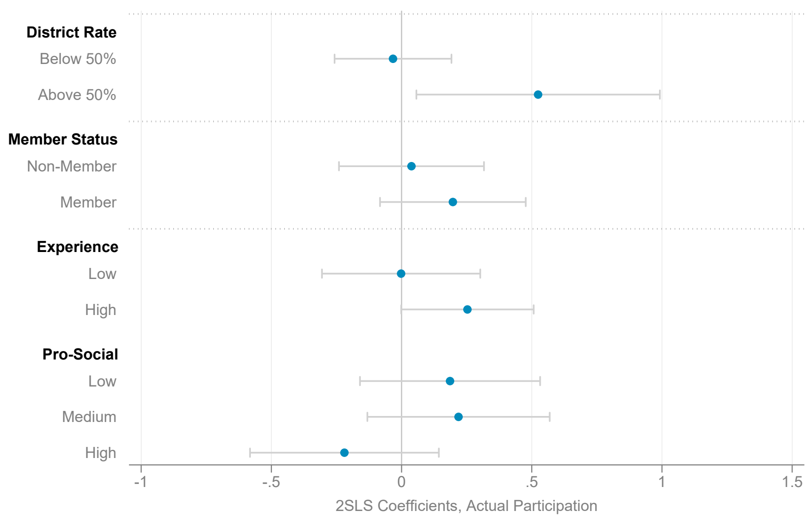
(b) Treatment Group

Note: The figure shows the distribution of priors (gray) and posteriors (blue) around the truthful estimate of the district membership rate. The district rate is normalized to 0. Panel A shows results for the control group; Panel B shows results for the treatment group. The x-axis is the difference in percentage points. The y-axis is the share of respondents in 5 percentage point bins.

Figure 4: Heterogeneity in 2SLS Estimates



(a) Planned Participation



(b) Actual Membership Status

Note: The figure plots two-stage least squares estimates from equation 3 by various subgroups. The outcome variable in panel A is planned participation in the union next year. The outcome variable in panel B is actual membership status according to the matching procedure described in Section II. The first set of results splits the sample by whether an individual's district membership rate is above or below 50 percent. The second set of results splits the sample by whether the respondent was a union member in the fall of 2023, before the randomization experiment. The third set of results splits the sample by whether the respondent had above (high) or below (low) median experience within the district. The last set of results splits the sample by whether the respondent received a low, medium, or high pro-sociality score. See Appendix Section B for details. The gray bars represent 95 percent confidence intervals.

Table 1: Selection into Survey Experiment

	(1) Non-Participants	(2) Participant Diff.	(3) P-value	(4) Observations
Experience	14.61	0.548***	0.01	56,837
		(0.193)		
FTE	99.56	0.834***	<0.01	56,837
		(0.203)		
Salary	64,729.19	-111.449	0.70	56,837
		(286.408)		
Number Positions	1.51	0.097***	<0.01	56,837
		(0.020)		
Advanced Degree	0.51	0.013	0.18	56,837
		(0.010)		
Reading Teacher	0.34	-0.012	0.21	56,837
		(0.009)		
Math Teacher	0.34	-0.033***	<0.01	56,837
		(0.009)		
Male	0.23	0.039***	<0.01	56,837
		(0.009)		
Non-White	0.07	-0.016***	<0.01	56,837
		(0.005)		
Member	0.37	0.057***	<0.01	56,837
		(0.010)		

Note: The sample is all teachers working in school districts that received the survey. Each row shows the results from regressing the stated baseline variable on an indicator variable for whether the individual took the survey. Column 1 presents the mean value for non-participants. Column 2 shows the coefficient and associated standard errors for the indicator denoting participation in the survey. Column 3 presents the associated p-value. The number of observations are in column 4. All regressions use district fixed effects and standard errors are clustered at the district level. *, **, *** denote statistical significance at the 10, 5, and 1 percent levels, respectively.

Table 2: Balance

	(1) Control	(2) Treated Diff.	(3) P-value	(4) Observations
Experience	15.08	0.069	0.86	2,492
		(0.387)		
FTE	100.51	-0.037	0.93	2,492
		(0.395)		
Salary	64,645.20	-173.169	0.75	2,492
		(558.565)		
Number Positions	1.60	0.017	0.66	2,492
		(0.040)		
Advanced Degree	0.52	0.009	0.64	2,492
		(0.020)		
Reading Teacher	0.33	-0.021	0.27	2,492
		(0.019)		
Math Teacher	0.29	0.017	0.35	2,492
		(0.018)		
Male	0.29	-0.034*	0.06	2,492
		(0.018)		
Non-White	0.06	-0.017*	0.06	2,492
		(0.009)		
Member	0.44	-0.020	0.31	2,492
		(0.020)		
Pro-Social	0.95	0.010	0.75	2,482
		(0.032)		
Prior Belief, District	45.55	-0.779	0.47	2,442
		(1.089)		
Prior Belief, School	44.16	-0.787	0.51	2,441
		(1.194)		
Attrited	0.03	-0.009	0.17	2,492
		(0.007)		

Note: Each row shows the results from regressing the stated baseline variable on treatment status. Column 1 presents the mean value in the control group. Column 2 shows the coefficient and associated standard errors on the treatment status indicator. Column 3 presents the associated p-value. The number of observations are in column 4. There are fewer observations for survey-based outcomes because some respondents did not fill out the associated question. *, **, *** denote statistical significance at the 10, 5, and 1 percent levels, respectively.

Table 3: First Stage Effect on Beliefs

	(1)	(2)	(3)
		Outcome = Posterior Beliefs	
Panel A: Pooled			
Treated (Recode)	8.589***	8.564***	8.712***
	(0.466)	(0.461)	(0.454)
Observations	2,422	2,422	2,422
Panel B: Underestimators			
Treated	5.094***	5.108***	5.343***
	(0.702)	(0.705)	(0.686)
Observations	1,080	1,080	1,080
Panel C: Overestimators			
Treated	-11.402***	-11.344***	-11.413***
	(0.613)	(0.599)	(0.595)
Observations	1,342	1,342	1,342
Baseline Membership	No	Yes	Yes
Additional Controls	No	No	Yes

Note: The outcome variable is respondents' estimates of the district membership rate next year. Panel A is the pooled sample of all participants, where treatment is recoded as -1 for those who overestimated their district share as described in Section IV. Panel B is the subsample of participants who underestimated their district share; Panel C is the subsample of participants who overestimated their district share. All specifications control for prior beliefs, column 2 controls for respondents' prior membership status, and column 3 includes additional controls as described in the text. The specifications in Panel A also include an indicator variable for one's prior beliefs being above the district membership rate and an interaction between the above district membership rate indicator and each control variable. Robust standard errors in parentheses. *, **, *** denote statistical significance at the 10, 5, and 1 percent levels, respectively.

Table 4: Reduced Form Effect on Planned Participation

	(1)	(2)	(3)
		Outcome = Planned Participation	
Panel A: Pooled			
Treated (Recode)	2.466	2.679**	3.160**
	(1.883)	(1.348)	(1.366)
Observations	2,422	2,422	2,422
Panel B: Underestimators			
Treated	0.116	1.729	2.339
	(2.689)	(1.906)	(1.915)
Observations	1,080	1,080	1,080
Panel C: Overestimators			
Treated	-4.357*	-3.444*	-3.818**
	(2.620)	(1.889)	(1.923)
Observations	1,342	1,342	1,342
Baseline Membership	No	Yes	Yes
Additional Controls	No	No	Yes

Note: The outcome variable is an indicator variable equal to 100 if the respondent said they were "very likely" or "somewhat likely" to be a member of the teachers' union next year. Panel A is the pooled sample of all participants, where treatment is recoded as -1 for those who overestimated their district share as described in Section IV. Panel B is the subsample of participants who underestimated their district share; Panel C is the subsample of participants who overestimated their district share. Column 1 includes controls for prior beliefs, column 2 adds a control for respondents' prior membership status, and column 3 includes additional controls as described in the text. The specifications in Panel A also include an indicator variable for one's prior beliefs being above the district membership rate and an interaction between the above district membership rate indicator and each control variable. Robust standard errors in parentheses. *, **, *** denote statistical significance at the 10, 5, and 1 percent levels, respectively.

Table 5: 2SLS Effect of Change in Posteriors on Planned Participation

	(1)	(2)	(3)
	Outcome = Planned Participation		
Panel A: Pooled			
Posterior Beliefs	0.287	0.313**	0.363**
	(0.216)	(0.155)	(0.153)
Observations	2,422	2,422	2,422
Panel B: Underestimators			
Posterior Beliefs	0.023	0.338	0.438
	(0.527)	(0.368)	(0.350)
Observations	1,080	1,080	1,080
Panel C: Overestimators			
Posterior Beliefs	0.382*	0.304*	0.334**
	(0.225)	(0.164)	(0.165)
Observations	1,342	1,342	1,342
Baseline Membership	No	Yes	Yes
Additional Controls	No	No	Yes

Note: The outcome variable is an indicator variable equal to 100 if the respondent said they were "very likely" or "somewhat likely" to be a member of the teachers' union next year. Panel A is the pooled sample of all participants, where treatment is recoded as -1 for those who overestimated their district share as described in Section IV. Panel B is the subsample of participants who underestimated their district share; Panel C is the subsample of participants who overestimated their district share. Column 1 includes controls for prior beliefs, column 2 adds a control for respondents' prior membership status, and column 3 includes additional controls as described in the text. The specifications in Panel A also include an indicator variable for one's prior beliefs being above the district membership rate and an interaction between the above district membership rate indicator and each control variable. Robust standard errors in parentheses. *, **, *** denote statistical significance at the 10, 5, and 1 percent levels, respectively.

Table 6: Reduced Form Effect on Actual Union Membership, One Year Later

	(1)	(2)	(3)
		Outcome = Union Member, 2024–25	
Panel A: Pooled			
Treated (Recode)	0.512	0.931	0.934
	(1.938)	(0.931)	(0.942)
Observations	2,255	2,255	2,255
Panel B: Underestimators			
Treated	-0.334	1.458	1.606
	(2.725)	(1.369)	(1.411)
Observations	1,003	1,003	1,003
Panel C: Overestimators			
Treated	-1.190	-0.509	-0.398
	(2.726)	(1.270)	(1.264)
Observations	1,252	1,252	1,252
Baseline Membership	No	Yes	Yes
Additional Controls	No	No	Yes

Note: The outcome variable is an indicator variable equal to 100 if the respondent was a union member in the 2024–25 school year according to the matching procedure described in Section II. Panel A is the pooled sample of all participants, where treatment is recoded as -1 for those who overestimated their district share as described in Section IV. Panel B is the subsample of participants who underestimated their district share; Panel C is the subsample of participants who overestimated their district share. Column 1 includes controls for prior beliefs, column 2 adds a control for respondents' prior membership status, and column 3 includes additional controls as described in the text. The specifications in Panel A also include an indicator variable for one's prior beliefs being above the district membership rate and an interaction between the above district membership rate indicator and each control variable. There are fewer observations than in Table 4 because I exclude individuals who left the teaching workforce or who moved to non-teaching roles. Robust standard errors in parentheses. *, **, *** denote statistical significance at the 10, 5, and 1 percent levels, respectively.

Table 7: 2SLS Effect of Change in Posteriors on Actual Union Membership, One Year Later

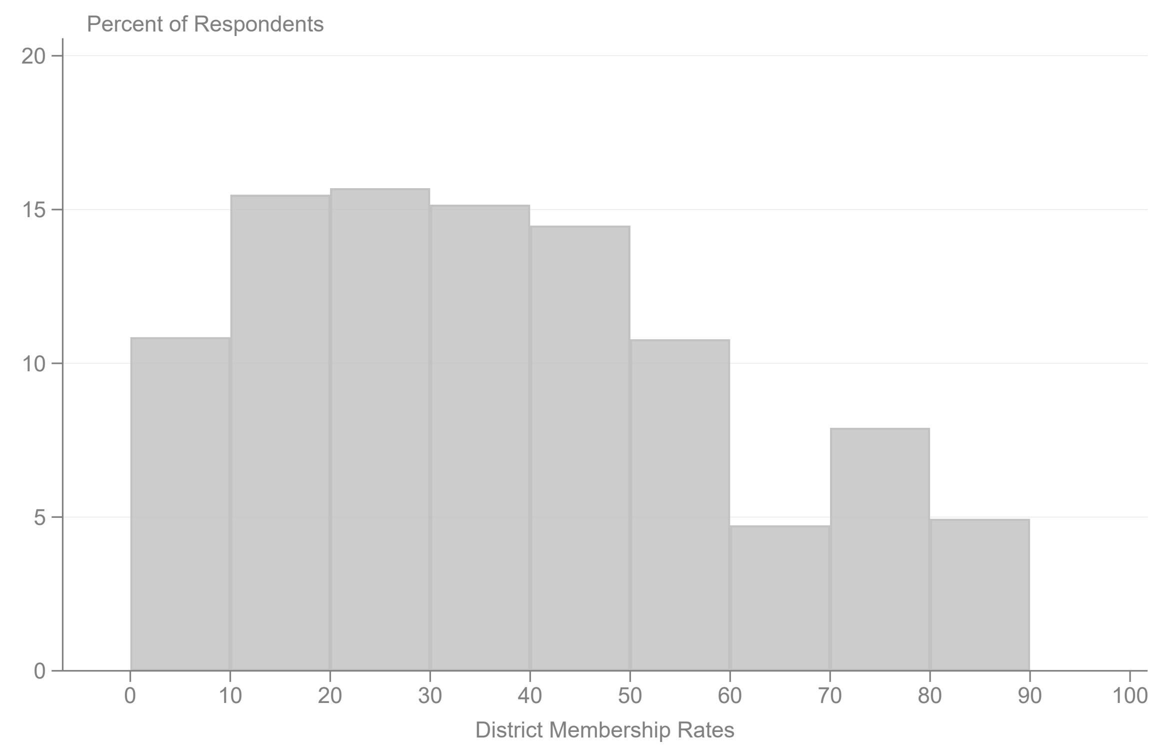
	(1)	(2)	(3)
		Outcome = Union Member, 2024–25	
Panel A: Pooled			
Posterior Beliefs	0.059	0.108	0.106
	(0.224)	(0.108)	(0.106)
Observations	2,255	2,255	2,255
Panel B: Underestimators			
Posterior Beliefs	-0.066	0.287	0.299
	(0.538)	(0.267)	(0.258)
Observations	1,003	1,003	1,003
Panel C: Overestimators			
Posterior Beliefs	0.104	0.044	0.034
	(0.236)	(0.111)	(0.108)
Observations	1,252	1,252	1,252
Baseline Membership	No	Yes	Yes
Additional Controls	No	No	Yes

Note: The outcome variable is an indicator variable equal to 100 if the respondent was a union member in the 2024–25 school year according to the matching procedure described in Section II. Panel A is the pooled sample of all participants, where treatment is recoded as -1 for those who overestimated their district share as described in Section IV. Panel B is the subsample of participants who underestimated their district share; Panel C is the subsample of participants who overestimated their district share. Column 1 includes controls for prior beliefs, column 2 adds a control for respondents' prior membership status, and column 3 includes additional controls as described in the text. The specifications in Panel A also include an indicator variable for one's prior beliefs being above the district membership rate and an interaction between the above district membership rate indicator and each control variable. There are fewer observations than in Table 5 because I exclude individuals who left the teaching workforce or who moved to non-teaching roles. Robust standard errors in parentheses. *, **, *** denote statistical significance at the 10, 5, and 1 percent levels, respectively.

Supplemental Appendix

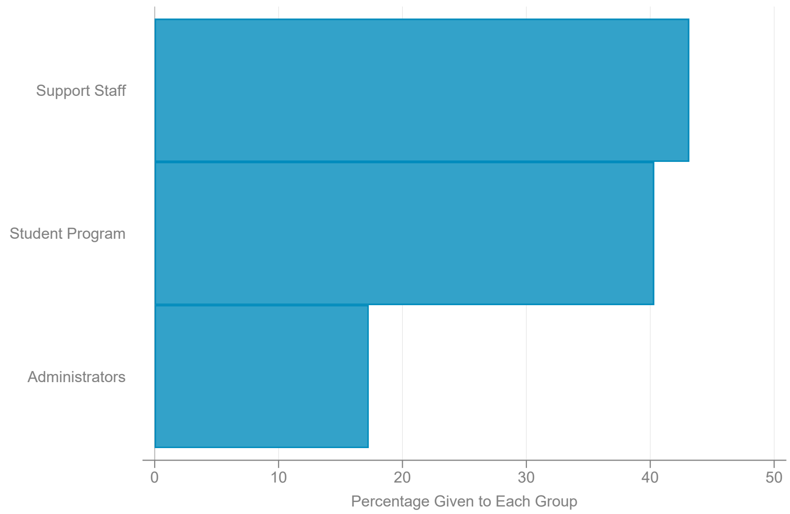
A. Additional Figures/Tables

Figure A1: Distribution of District Membership Rates

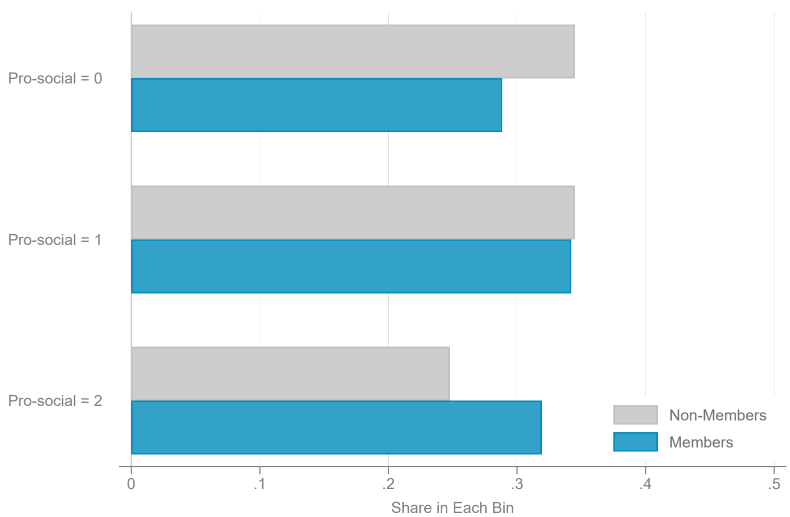


Note: The x-axis is the district membership rate that was used in the experimental information provision. The y-axis shows the percentage of survey participants in each 10 percentage point bin.

Figure A2: Descriptive Statistics on Pro-Social Measure



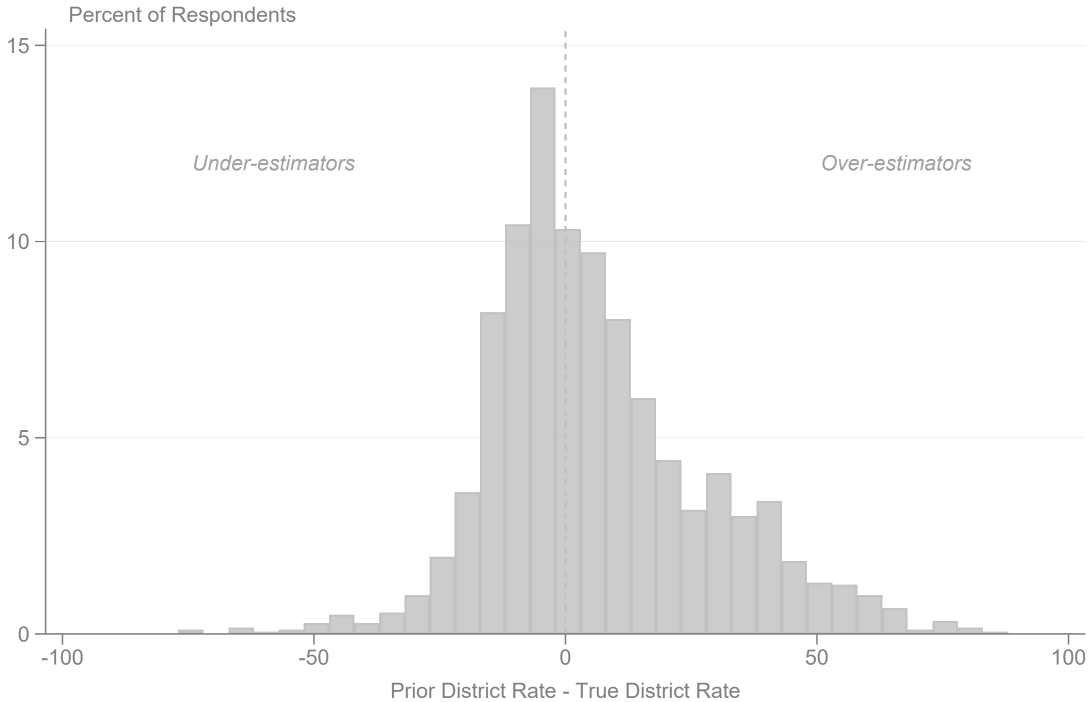
(a) Average Share Given to Each Group



(b) Pro-Social Score by Union Status

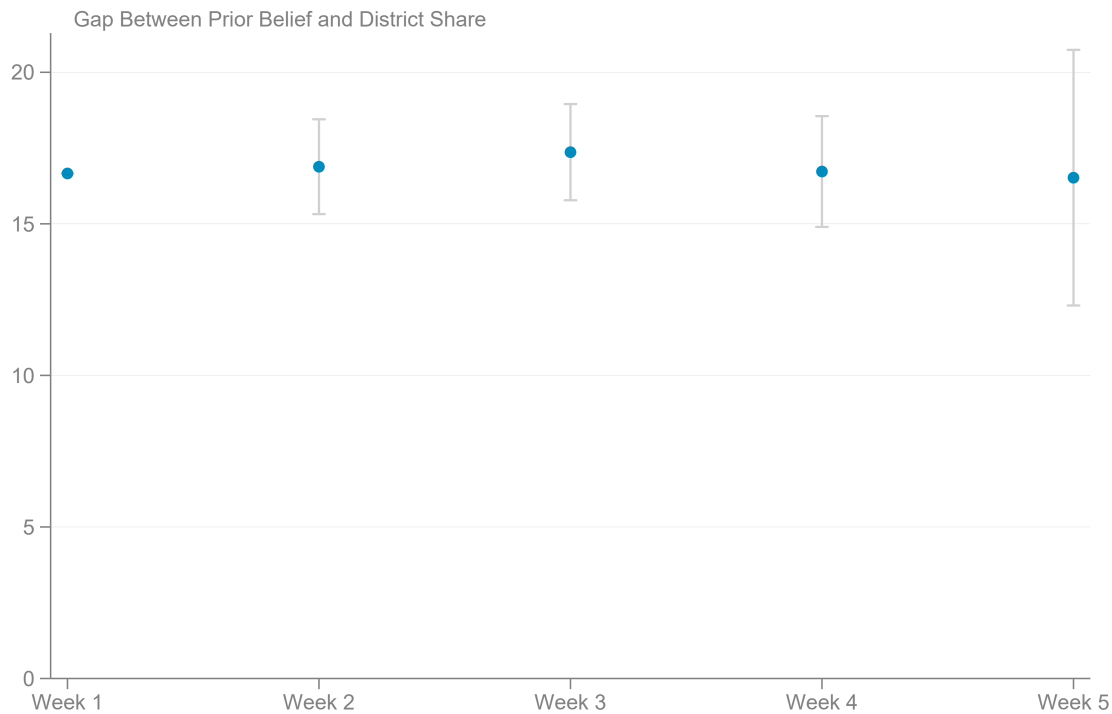
Note: Panel A shows the average percentage of the budget surplus shared between support staff pay, student program funding, and administrator pay, respectively. See Appendix Section C. Panel B shows the share of respondents within each pro-social score separated by union status.

Figure A3: Distribution of Priors Excluding Districts Near 50 Percent



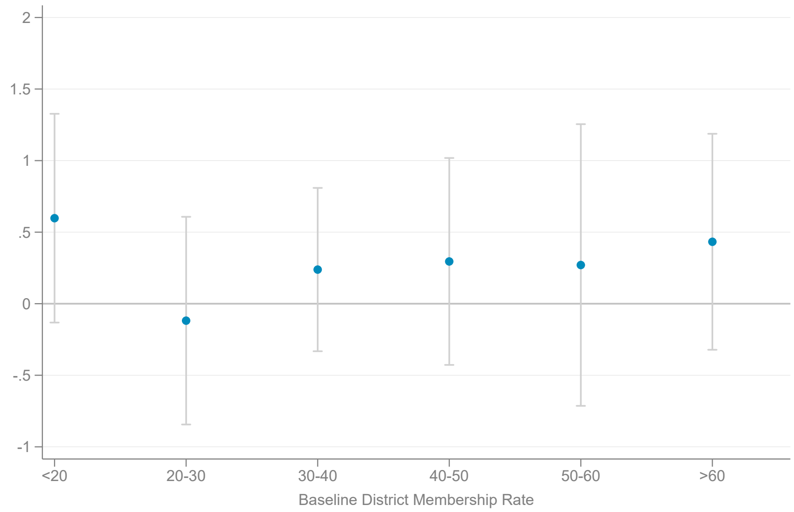
Note: This is a reproduction of Figure 2, but restricted to districts where the estimated membership rate was less than 40 percent or greater than 60 percent. The x-axis is an individual's prior belief of their school district's membership rate minus the truthful estimate of the district membership rate. The y-axis is the share of respondents who fall into each 5 percentage point bin. Respondents who underestimated their district membership rate lie to the left of the dashed line; respondents who overestimated their district rate lie to the right of the dashed line.

Figure A4: Gap Between Prior Beliefs and District Rate by Survey Week

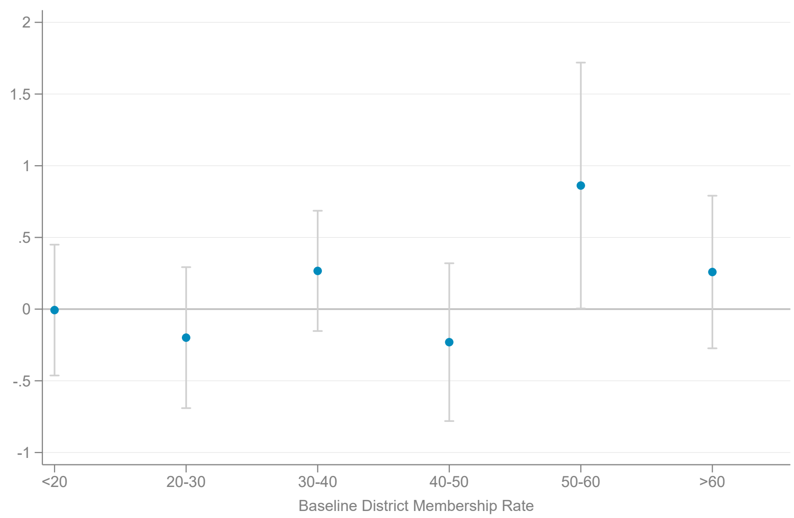


Note: This figure presents the gap between prior beliefs and the district membership rate regressed on indicators for which week the survey was taken. The coefficients from this regression are then added to the Week 1 mean of 16.6 percentage points. Gray bars represent 95 percent confidence intervals.

Figure A5: Effects by 10 Percentage Point Bins of District Membership Share



(a) Planned Participation



(b) Actual Participation

Note: Both figures show the 2SLS results broken down by 10 percentage point bins of the pre-experimental district membership rates. This figure uses the parsimonious specification from Table 5 and Table 7 column 2, due to the smaller sample size within each bin. I pool respondents with district membership rates below 20 percent, as there are few districts with membership rates below 10 percent. Likewise, I pool all respondents with membership rates greater than 60 percent, as there are few districts with membership rates greater than 70 percent. The outcome variables are planned participation (panel A) and actual participation (panel B). Gray bars represent 95 percent confidence intervals.

Table A1: Balance, Underestimators

	(1) Control	(2) Treated Diff.	(3) P-value	(4) Observations
Experience	14.43	0.575	0.33	1,091
		(0.587)		
FTE	100.31	-0.291	0.65	1,091
		(0.635)		
Salary	63,933.05	131.643	0.88	1,091
		(846.189)		
Number Positions	1.59	0.053	0.41	1,091
		(0.065)		
Advanced Degree	0.48	0.019	0.52	1,091
		(0.030)		
Reading Teacher	0.39	-0.050*	0.09	1,091
		(0.029)		
Math Teacher	0.33	0.016	0.56	1,091
		(0.029)		
Male	0.26	-0.014	0.59	1,091
		(0.027)		
Non-White	0.07	-0.010	0.51	1,091
		(0.015)		
Member	0.44	-0.032	0.29	1,091
		(0.030)		
Pro-Social	0.94	0.013	0.79	1,087
		(0.048)		
Prior Belief, District	28.94	-0.892	0.54	1,091
		(1.470)		
Prior Belief, School	29.43	-1.096	0.52	1,091
		(1.686)		
Attrited	0.01	-0.002	0.76	1,091
		(0.006)		

Note: The sample is respondents whose prior belief was below or equal to their respective district share. Each row shows the results from regressing the stated baseline variable on treatment status. Column 1 presents the mean value in the control group. Column 2 shows the coefficient and associated standard errors on the treatment status indicator. Column 3 presents the associated p-value. The number of observations are in column 4. *, **, *** denote statistical significance at the 10, 5, and 1 percent levels, respectively.

Table A2: Balance, Overestimators

	(1) Control	(2) Treated Diff.	(3) P-value	(4) Observations
Experience	15.85	-0.409	0.43	1,351
		(0.522)		
FTE	100.63	0.202	0.68	1,351
		(0.494)		
Salary	65,506.13	-527.069	0.48	1,351
		(755.039)		
Number Positions	1.61	-0.018	0.72	1,351
		(0.052)		
Advanced Degree	0.55	0.007	0.80	1,351
		(0.027)		
Reading Teacher	0.29	0.000	0.99	1,351
		(0.025)		
Math Teacher	0.27	0.007	0.77	1,351
		(0.024)		
Male	0.31	-0.047*	0.05	1,351
		(0.025)		
Non-White	0.06	-0.021*	0.08	1,351
		(0.012)		
Member	0.46	-0.016	0.56	1,351
		(0.027)		
Pro-Social	0.94	0.008	0.85	1,345
		(0.043)		
Prior Belief, District	58.72	-0.195	0.86	1,351
		(1.124)		
Prior Belief, School	55.86	-0.114	0.93	1,350
		(1.359)		
Attrited	0.01	-0.001	0.77	1,351
		(0.004)		

Note: The sample is respondents whose prior belief was above their respective district share. Each row shows the results from regressing the stated baseline variable on treatment status. Column 1 presents the mean value in the control group. Column 2 shows the coefficient and associated standard errors on the treatment status indicator. Column 3 presents the associated p-value. The number of observations are in column 4. *, **, *** denote statistical significance at the 10, 5, and 1 percent levels, respectively.

Table A3: Characteristics that Predict Informed Priors on the Union

	(1) Absolute Difference	(2) Overestimated	(3) Certification Status
Member	-1.846***	0.055**	0.099***
	(0.661)	(0.023)	(0.020)
High Experience	-2.408***	0.013	0.062***
	(0.613)	(0.021)	(0.018)
Assignment FTE	-0.097***	-0.000	-0.001
	(0.031)	(0.001)	(0.001)
Number Positions	-0.831***	-0.021**	0.023**
	(0.307)	(0.010)	(0.009)
Advanced Degree	-0.807	0.012	0.026
	(0.608)	(0.021)	(0.018)
Reading Teacher	-0.009	-0.038	-0.003
	(0.753)	(0.026)	(0.022)
Math Teacher	-0.736	-0.017	0.006
	(0.763)	(0.026)	(0.023)
Male	0.809	0.035	0.031
	(0.657)	(0.022)	(0.019)
Observations	2,440	2,440	2,440
Mean	16.91	0.55	0.76

Note: The table shows what characteristics predict prior beliefs about the local union. The outcome in column 1 is the absolute difference between one's prior belief and the district-specific membership rate. The outcome in column 2 is an indicator variable equal to one if the respondent overestimated their district membership rate. The outcome in column 3 is an indicator variable equal to one if the respondent correctly knew the certification status in their district. All columns include fixed effects for the district membership rate. *, **, *** denote statistical significance at the 10, 5, and 1 percent levels, respectively.

Table A4: Learning Rates

	Updating (Posterior - Prior)
Treated (Recode)	-0.508
	(0.683)
Signal Gap	-0.006
	(0.040)
Treated (Recode) × Signal Gap	0.547***
	(0.043)
Observations	2,422

Note: The table estimates the degree of learning for members of the treatment group. The outcome variable is the difference between a respondent's posterior and prior beliefs. Treated (Recode) is an indicator for whether the respondent was in the treatment group, where overestimators are coded as -1 (see equation 1). The "signal gap" is the difference between the district membership rate and one's prior beliefs. The third row shows the interaction term between these two variables. Robust standard errors are in parentheses. *, **, *** denote statistical significance at the 10, 5, and 1 percent levels, respectively.

Table A5: Reduced Form Effect on Likelihood of Participation

	(1)	(2)	(3)
		Outcome = Likelihood Scale	
Panel A: Pooled			
Treated (Recode)	0.123*	0.131***	0.152***
	(0.066)	(0.046)	(0.046)
Observations	2,422	2,422	2,422
Panel B: Underestimators			
Treated	0.051	0.109	0.133**
	(0.095)	(0.067)	(0.067)
Observations	1,080	1,080	1,080
Panel C: Overestimators			
Treated	-0.181**	-0.149**	-0.168***
	(0.090)	(0.064)	(0.064)
Observations	1,342	1,342	1,342
Baseline Membership	No	Yes	Yes
Additional Controls	No	No	Yes

Note: The outcome variable is a 5-point scale of the likelihood that a respondent is a member of the teachers' union next year where 5 = very likely and 1 = very unlikely. Panel A is the pooled sample of all participants, where treatment is recoded as -1 for those who overestimated their district share as described in Section IV. Panel B is the subsample of participants who underestimated their district share; Panel C is the subsample of participants who overestimated their district share. Column 1 includes controls for prior beliefs, column 2 adds a control for respondents' prior membership status, and column 3 includes additional controls as described in the text. The specifications in Panel A also include an indicator variable for one's prior beliefs being above the district membership rate and an interaction between the above district membership rate indicator and each control variable. Robust standard errors in parentheses. *, **, *** denote statistical significance at the 10, 5, and 1 percent levels, respectively.

Table A6: Effect of Change in Others' Membership on Own Membership, Admin Data

	Outcome = Membership Status	
Δ District Membership Rate	0.369***	0.365***
	(0.021)	(0.020)
Lagged Membership	0.884***	0.878***
	(0.005)	(0.005)
Observations	310,909	310,909
Controls	No	Yes
Number Districts	405	405

Note: The table estimates the relationship between the leave-one-out change in district membership rates and a teacher's membership status. The sample is teachers in the administrative data from 2017–2022. The outcome variable is a teacher's membership status in a given year t. The variable of interest, Δ District Membership Rate, is the change in the leave-one-out district membership rates in the school district where a teacher worked. Lagged membership is an indicator variable for a teacher's membership status in the prior school year. Both specifications include district and year fixed effects. Column 2 includes additional controls for local experience, full-time employment amount, the number of positions a teacher worked, higher education, indicators for teaching reading or math, and indicators for being male or non-white. Standard errors clustered at the district level are in parentheses. *, **, *** denote statistical significance at the 10, 5, and 1 percent levels, respectively.

Table A7: Robustness Exercises, Planned Participation

	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: First Stage (Posterior Beliefs)						
Treated (Recode)	8.712***	7.602***	8.945***	8.814***	8.644***	8.368***
	(0.454)	(0.468)	(0.485)	(0.452)	(0.454)	(0.440)
Panel B: Reduced Form (Planned Participation)						
Treated (Recode)	3.160**	3.054**	4.133***	3.179**	3.107**	2.982**
	(1.366)	(1.358)	(1.452)	(1.370)	(1.365)	(1.359)
Panel C: 2SLS (Planned Participation)						
Posterior Beliefs	0.363**	0.402**	0.462***	0.361**	0.359**	0.356**
	(0.153)	(0.174)	(0.156)	(0.152)	(0.154)	(0.159)
Observations	2,422	2,420	2,386	2,422	2,361	2,397
School Share	No	Yes	No	No	No	No
District FEs	No	No	Yes	No	No	No
Recode Exacts	No	No	No	Yes	No	No
Drop Attention Check	No	No	No	No	Yes	No
Drop Outliers	No	No	No	No	No	Yes

Note: This table reproduces the main results with alternative specifications or samples. Panel A presents the first stage results on posterior beliefs, panel B presents the reduced form effect on planned participation, and panel C presents the 2SLS results of a change in posteriors on planned participation. Column 1 reproduces the main specification as in column 3 of Tables 3–5. Column 2 replaces the district-level priors and posteriors with the corresponding prior and posterior beliefs about union membership rates within a teacher's school. Column 3 includes district fixed effects and clusters standard errors at the district level. Column 4 recodes the people who guessed the district share perfectly from 1 to 0. Column 5 drops those respondents who failed the attention check. Column 6 drops those observations for which the gap between the district share and their prior belief was greater than the 99th percentile. All specifications use robust standard errors and column 3 clusters standard errors at the district level. *, **, *** denote statistical significance at the 10, 5, and 1 percent levels, respectively.

Table A8: 2SLS Effect of Change in Posteriors on Planned Participation, Dropping Attrition Group

	(1)	(2)	(3)
		Outcome = Planned Participation	
Panel A: Pooled			
Posterior Beliefs	0.358	0.405***	0.445***
	(0.222)	(0.155)	(0.153)
Observations	2,252	2,252	2,252
Panel B: Underestimators			
Posterior Beliefs	0.103	0.441	0.525
	(0.548)	(0.371)	(0.353)
Observations	1,002	1,002	1,002
Panel C: Overestimators			
Posterior Beliefs	0.448*	0.403**	0.416**
	(0.230)	(0.163)	(0.164)
Observations	1,250	1,250	1,250
Baseline Membership	No	Yes	Yes
Additional Controls	No	No	Yes

Note: The outcome variable is an indicator variable equal to 100 if the respondent said they were "very likely" or "somewhat likely" to be a member of the teachers' union next year. Panel A is the pooled sample of all participants, where treatment is recoded as -1 for those who overestimated their district share as described in Section IV. Panel B is the subsample of participants who underestimated their district share; Panel C is the subsample of participants who overestimated their district share. Column 1 includes controls for prior beliefs, column 2 adds a control for respondents' prior membership status, and column 3 includes additional controls as described in the text. The specifications in Panel A also include an indicator variable for one's prior beliefs being above the district membership rate and an interaction between the above district membership rate indicator and each control variable. Robust standard errors in parentheses. *, **, *** denote statistical significance at the 10, 5, and 1 percent levels, respectively.

Table A9: 2SLS Effect of Change in Posteriors on Actual Union Membership, One Year Later (≥ 50% Districts)

	(1)	(2)	(3)
		Outcome = Union Member, 2024–25	
Panel A: Pooled			
Posterior Beliefs	0.749*	0.503**	0.524**
	(0.448)	(0.236)	(0.239)
Observations	679	679	679
Panel B: Underestimators			
Posterior Beliefs	0.501	0.547*	0.573**
	(0.540)	(0.279)	(0.281)
Observations	353	353	353
Panel C: Overestimators			
Posterior Beliefs	1.113	0.436	0.447
	(0.770)	(0.415)	(0.427)
Observations	326	326	326
Baseline Membership	No	Yes	Yes
Additional Controls	No	No	Yes

Note: The sample is all districts with ≥50 percent baseline membership. The outcome variable is an indicator variable equal to 100 if the respondent was a union member in the 2024–25 school year according to the matching procedure described in Section II. Panel A is the pooled sample of all participants, where treatment is recoded as -1 for those who overestimated their district share as described in Section IV. Panel B is the subsample of participants who underestimated their district share; Panel C is the subsample of participants who overestimated their district share. Column 1 includes controls for prior beliefs, column 2 adds a control for respondents' prior membership status, and column 3 includes additional controls as described in the text. The specifications in Panel A also include an indicator variable for one's prior beliefs being above the district membership rate and an interaction between the above district membership rate indicator and each control variable. Robust standard errors in parentheses. *, **, *** denote statistical significance at the 10, 5, and 1 percent levels, respectively.

Table A10: 2SLS Effect of Change in Posteriors on Actual Union Membership, One Year Later (< 50% Districts)

	(1)	(2)	(3)
		Outcome = Union Member, 2024–25	
Panel A: Pooled			
Posterior Beliefs	-0.147	-0.013	-0.033
	(0.250)	(0.121)	(0.115)
Observations	1,576	1,576	1,576
Panel B: Underestimators			
Posterior Beliefs	-0.686	-0.028	-0.094
	(1.015)	(0.503)	(0.447)
Observations	650	650	650
Panel C: Overestimators			
Posterior Beliefs	-0.050	-0.011	-0.021
	(0.235)	(0.110)	(0.106)
Observations	926	926	926
Baseline Membership	No	Yes	Yes
Additional Controls	No	No	Yes

Note: The sample is all districts with <50 percent baseline membership. The outcome variable is an indicator variable equal to 100 if the respondent was a union member in the 2024–25 school year according to the matching procedure described in Section II. Panel A is the pooled sample of all participants, where treatment is recoded as -1 for those who overestimated their district share as described in Section IV. Panel B is the subsample of participants who underestimated their district share; Panel C is the subsample of participants who overestimated their district share. Column 1 includes controls for prior beliefs, column 2 adds a control for respondents' prior membership status, and column 3 includes additional controls as described in the text. The specifications in Panel A also include an indicator variable for one's prior beliefs being above the district membership rate and an interaction between the above district membership rate indicator and each control variable. Robust standard errors in parentheses. *, **, *** denote statistical significance at the 10, 5, and 1 percent levels, respectively.

B. Data Appendix

A. Measuring Union Membership

To measure union membership, I link personnel data from the Wisconsin Department of Public Instruction to campaign finance reports from the Wisconsin Campaign Finance Information System. For the campaign contributions data, I use contributions to the state teachers' union (WEAC PAC) and any of its regional affiliates. The unique information common to both datasets is a person's full name and the regional WEAC chapter they work for in a given year. For the DPI data, this information is linked via district identifiers from here: https://weac.org/region-finder-test/.

In the DPI data, 99 percent of people are uniquely identified by name, year, and WEAC region. In the experimental sample, there are 25 individuals who are not uniquely identified by name, year, and WEAC region. For these duplicates, there are three possible scenarios: (1) The number of duplicates matches the number of unique people in the WCFIS data. (2) No one appears in the WCFIS data. (3) There is one or more people in the WCFIS data, but it is not immediately clear which person matches which teacher. In the first case, all of the duplicates are considered members, while in the second, no one is a member. In the third case, I denote membership status by which teacher's address is closest to their respective school. There are six observations that meet this latter criteria.

The campaign contributions data is updated at least once every quarter of the year. I denote someone as a member before the experiment if they are in the contribution data in the fall of 2023 or January 2024. The survey went out in the spring of 2024, meaning that the later reports were potentially affected by the survey and treatment.

B. Experimental Analysis Sample

There were 2,789 individuals who filled out at least one question of the survey and 2,525 who completed enough of the survey to be included in the experimental randomization. In the analysis sample, I drop 20 respondents who were not licensed teachers as support staff often have no union representation or different unions than licensed teaching staff. I also drop 13 people who could not be matched to the administrative personnel data. In the main analysis, I also exclude 70 people who did not answer any of the three questions on prior beliefs, posterior beliefs, or planned participation.

C. Data Variables

Advanced degree: An indicator variable equal to 1 if the teacher had a master's degree or higher.

Attrited: An indicator variable equal to 1 if the respondent was randomized into either treatment or control but did not answer one or more of the main questions relating to prior beliefs, posterior beliefs, or planned participation.

Baseline member: An indicator variable equal to 1 if the teacher was a union member at baseline according to the campaign finance reports. I consider teachers as members at baseline if they appear in the fall or January reports from the 2023–24 school year. (The January reports cite that the contributions are from August to December 2023).

Experience: The number of years that a teacher has worked in their local school district.

FTE: A number corresponding to a teacher's full-time employment status, where 100 equals full-time.

Male: An indicator variable equal to 1 if the teacher was coded as male.

Math teacher: An indicator variable equal to 1 if the teacher worked as a math teacher. Teachers who teach multiple subjects, such as elementary teachers, are included.

Non-white: An indicator variable equal to 1 if the teacher's race in the administrative data was not "white."

Number positions: The number of positions that a teacher worked. For instance, a teacher may work part-time in two different schools.

Planned participation: An indicator variable equal to 1 if the respondent answered "very likely" or "somewhat likely" to the question regarding the likelihood of being in the union next year.

Posterior belief, district: A number ranging from 0 to 100 of a teacher's estimate of the union membership rate in their school district next school year (2024–25). See Appendix Section C for a screenshot of the posterior belief question.

Posterior belief, school: A number ranging from 0 to 100 of a teacher's estimate of the union membership rate in their school next school year (2024–25). See Appendix Section C for a screenshot of the posterior belief question.

Prior belief, district: A number ranging from 0 to 100 of a teacher's estimate of the union membership rate in their school district last school year (2022–23). See Appendix Section C for a screenshot of the prior belief question.

Prior belief, school: A number ranging from 0 to 100 of a teacher's estimate of the union membership rate in their school last school year (2022–23). See Appendix Section C for a screenshot of the prior belief question.

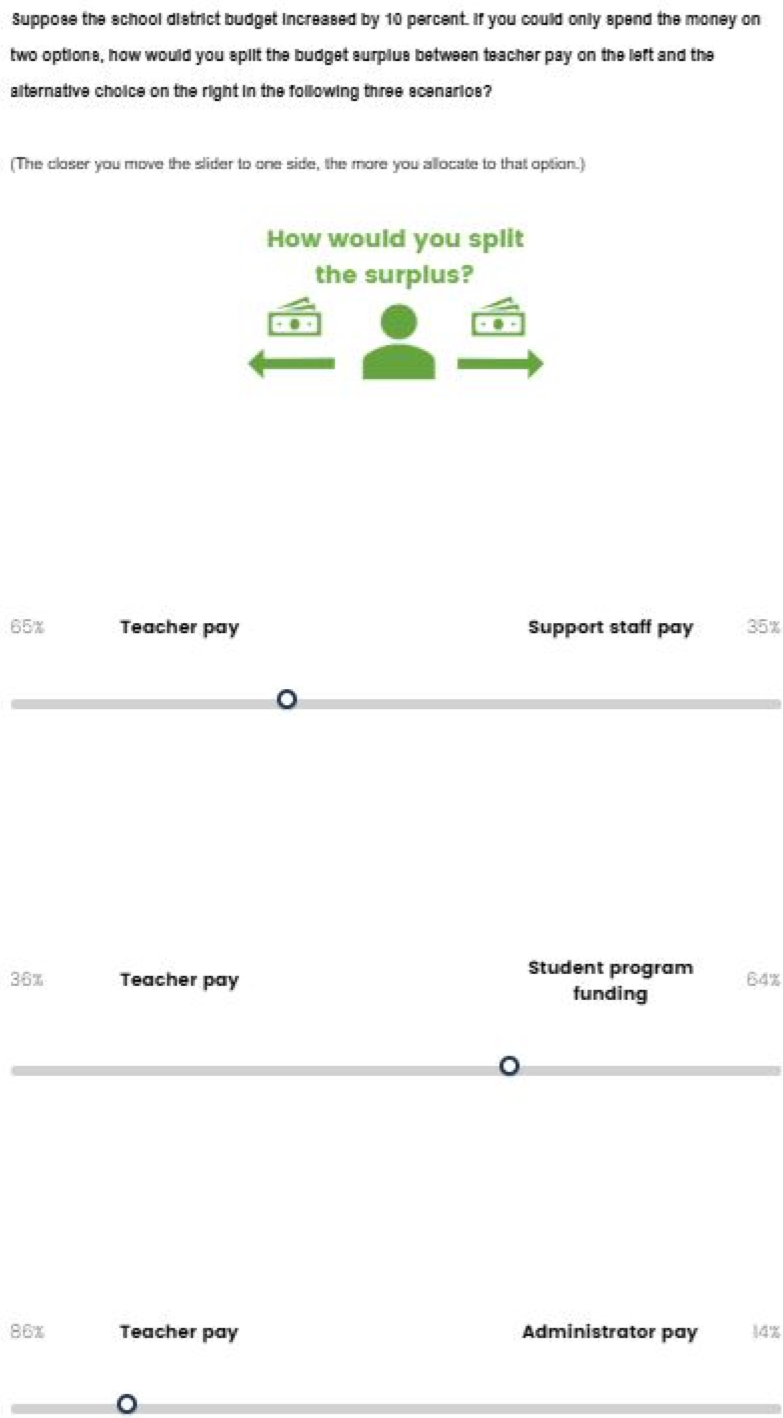
Pro-social: A variable from 0–2 indicating a survey participant's pro-sociality. The score is determined by whether the respondent shared at least 50 percent of the hypothetical budget surplus with any group. The score equals 1 if they shared 50 percent with one group and 2 if they shared 50 percent with at least two groups. See Appendix Section C for a screenshot of the pro-sociality question.

Reading teacher: An indicator variable equal to 1 if the teacher worked as a reading or English and language arts teacher. Teachers who teach multiple subjects, such as elementary teachers, are included.

Salary: A teacher's gross salary in dollars.

C. Survey Details

Below, I display screenshots of the Qualtrics survey questions used in the paper. Before randomization, I asked questions designed to proxy for teachers' pro-sociality. The following screenshot shows this, in which survey respondents moved the sliders to allocate a budget surplus in three hypothetical scenarios (Enke, Rodríguez-Padilla, and Zimmermann 2023).


Suppose the school district budget increased by 10 percent. If you could only spend the money on two options, how would you split the budget surplus between teacher pay on the left and the alternative choice on the right in the following three scenarios?

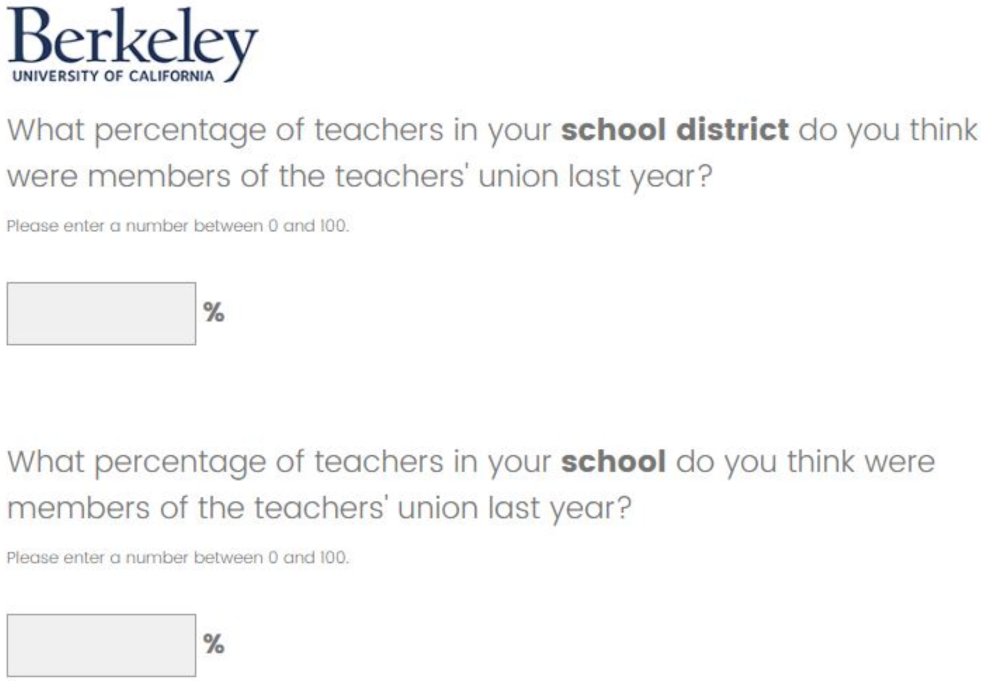
(The closer you move the slider to one side, the more you allocate to that option.)

65% Teacher pay Support staff pay 35%

36% Teacher pay Student program funding 64%

86% Teacher pay Administrator pay 14%

The next screenshot shows the question used to elicit prior beliefs about the district and school membership rates:


Berkeley
UNIVERSITY OF CALIFORNIA

What percentage of teachers in your **school district** do you think were members of the teachers' union last year?

Please enter a number between 0 and 100.

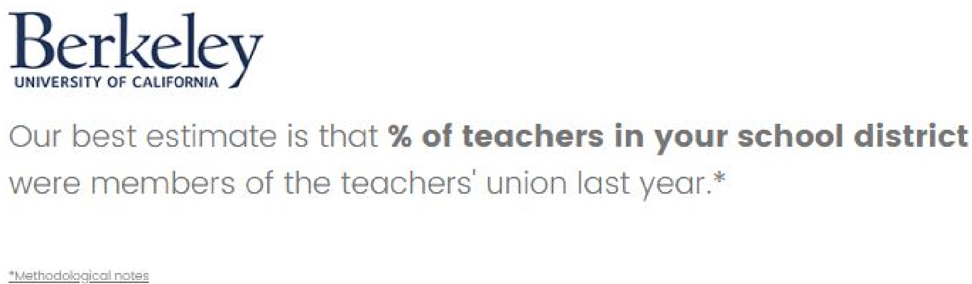
%

What percentage of teachers in your **school** do you think were members of the teachers' union last year?

Please enter a number between 0 and 100.

%

After eliciting prior beliefs, the treatment group was presented with the information provision seen below (control group teachers did not see this screen).


Berkeley
UNIVERSITY OF CALIFORNIA

Our best estimate is that **% of teachers in your school district** were members of the teachers' union last year.*

*Methodological notes

Before the "%" sign was a number that corresponded to the district-specific membership rate. If one clicked on the "Methodological notes" hyperlink, they saw the text: "Chapter 11 of Wisconsin state statute requires that political action committees itemize the source of all funds received as part of campaign finance reporting. The Wisconsin Campaign Finance Information System makes these reports publicly available semi-annually. The Wisconsin Educational Association Council sends a small share of annual membership dues to the WEAC political action committee, unless a member requests in writing a refund of the PAC contribution. Therefore, the estimate of a district's membership rate is constructed by linking the WEAC PAC listings with publicly available data from the Wisconsin Department of Public Instruction " 'All Staff Report'."

Next, all respondents (both treated and control) saw the following questions to elicit posterior beliefs and planned union participation in the next school year:

What percentage of teachers in your **school district** do you think will be members next year?

Please enter a number between 0 and 100.

%

What percentage of teachers in your **school** do you think will be members next year?

Please enter a number between 0 and 100.

%

How likely is it that you will be a member of the teachers' union next year?

- Very likely
- Somewhat likely
- Undecided
- Somewhat unlikely
- Very unlikely